Autonomous Learning of the Robot Kinematic Model

Alberto Dalla Libera, Nicola Castaman, Stefano Ghidoni, and Ruggero Carli

***Abstract*—Robotics systems are becoming more and more autonomous and reconfigurable. In this context, the design of algorithms capable of deriving kinematics and dynamics models directly from data could be particularly useful. In this article, we present an algorithm that learns a forward kinematics model of a robot starting from a time series of visual observations. Our strategy can be applied to any robot with serial kinematics composed of revolute and prismatics joints. First, the algorithm identifies the robot kinematic structure, i.e., a high-level description of the robot geometry that defines the connections between the rigid-bodies composing the robot. Then, the algorithm derives the forward kinematics relying on a Gaussian process (GP) model. More precisely, the GP model is based on a polynomial kernel, defined exploiting the kinematic structure previously identified. The effectiveness of the proposed solution has been tested via extensive Monte Carlo simulations, as well as via experiments on a real UR10 robot.**

***Index Terms*—Calibration and identification, kinematics, learning and adaptive systems, modular robots.**

I. Introduction

KINEMATIC models are of paramount importance in robotics. Several control strategies and trajectory planning algorithms are designed relying on the knowledge of a forward kinematics model, see [1], [2]. When dealing with a robot manipulator, forward kinematics models describe the map between joint configurations and links poses. If the geometrical parameters of the manipulator are known, first-principle models of this map are derived defining a proper sequence of consecutive roto-translations, that, typically, is parametrized adopting the Denavit–Hartenberg (DH) convention.

Typically, the robot geometrical parameters are accurately known, since most of the times CAD models are available, and direct measurements of the robot parameters are possible. However, robots are becoming more and more autonomous and reconfigurable, and in some cases such information about geometry is not available *a priori*. Consider for instance modular robotics [3]–[7], where robots are composed of different elementary building blocks, which can be interchanged, added or removed. By changing their geometry, modular robots adapt to different environments and tasks. Another relevant application where robot's geometry might change during the execution of a task is automatic adaptation to hardware failures. Hardware failures could result in an unexpected variation of the robot kinematic; consider, for instance, the extreme case of a robot joint getting stuck. In some cases, the robot might still be able to accomplish the task, by updating its model, and in turn, adapting its controller.

In the scenarios outlined above, where prior knowledge about robot geometry is partial or missing, standard solutions cannot be applied to derive kinematic models. Hence, the design of algorithms able to autonomously learn a geometrical model of a robot is essential to reduce set up times and human intervention, together with the associated costs. In such contexts, the identification of the forward kinematics is only a subtask needed to derive a robot geometrical model. The first step consists in identifying the kinematic structure, that is, a high-level description of the robot that defines how the different components are connected. In particular, we consider manipulator with serial kinematics. In this case, the kinematic structure determines the order by which links and joints occur in the kinematic chain, as well as the type of each joint, e.g., if a joint is revolute or prismatic. This high-level description is particularly useful in several robotic applications, like task and motion planning, transfer and meta learning [8].

The problem of identifying the robot geometry has been already addressed in the robotics community. The proposed approaches can be roughly divided based on the kind of sensors adopted to collect data. In [9], points cloud data have been used. First, links are identified and ordered by clustering points according to their relative distances and by assuming that each cluster corresponds to one link. After that, a forward kinematics model is identified optimizing the DH parameters with standard gradient descent. A more convenient setup has been considered in [10], [11], [12], and [13], where a distinct fiducial marker is attached to each link, making clustering operations not necessary. The robot kinematic structure is identified by learning the correct sequence of fiducial markers, as well as the type of connection between subsequent markers.

In particular, in [13], the kinematic structure and the forward kinematics are learned simultaneously. The solution is based on the identification of a Bayesian network, where the nodes of the network are random variables related to the markers

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poses and joint signals, while the edges represent the relative transformations between markers. In a first phase, adopting a Gaussian process regression (GPR) approach, a score is assigned to all the possible edges. Then, the kinematic model is identified by searching for the path that connects all the markers and minimizes a cost function properly defined.

In this article, we consider the same setup of [13] but, instead of learning simultaneously the kinematic structure and the forward kinematics, we propose a novel two-steps procedure. First, we identify the kinematic structure. Second, we learn a forward kinematics relying on GPR and exploiting the kinematic structure identified in the previous step. As far as the first step is concerned, the solution proposed is based on checking the feasibility of three systems of equations, obtained starting from standard kinematic relations between pairs of subsequent links. More precisely, given the measured poses of a couple of markers attached to subsequent links, and the corresponding joint input signal, a linear system of equations holds true if the three elements define a prismatic transformation; instead, a linear and a nonlinear system are satisfied if the transformation is revolute. Compared to the state of the art, the proposed approach is computationally much more efficient, being based on the solution of linear and nonlinear systems of equations with low dimensionality. Moreover, differently from [13], the proposed algorithm fully reconstructs the kinematic structure, including the joint type sequence.

The standard procedure applied to identify the forward kinematics consists in deriving a first-principle model based on the nominal parameters, and improving the model accuracy through calibration procedures, see for instance [14] [15]. However, in situations where the knowledge about nominal parameters is partial, or missing, this strategy might be not effective. Indeed, due to the nonlinearity of kinematic models, calibration procedures are effective only when deviations from nominal values are small. Hence, bad initialization of the kinematic parameters could significantly affect the performance of calibration algorithms. Given the small, or null, amount of prior information available in the setup considered, we propose an alternative solution that requires minimal prior information.

In this article, we propose a grey-box solution based on GPR. Starting from the knowledge of the kinematic structure, the standard inputs, i.e., the positions of the joints, are mapped into an augmented space where the forward kinematics derived according to DH convention is a polynomial function. As a consequence, each element of the output, namely, the pose of the marker attached to the last link, is modeled as a distinct GPR relying on a proper polynomial kernel. Experimental results show the effectiveness of our solution. Moreover, to further test the accuracy of the derived model, we have implemented a simple kinematic controller based on the learned model. It is worth mentioning that our work significantly extends the seminal intuitions explored in [16], where the authors have experimented the use of neural networks, comparing performance of different architectures via experiments performed on a simulated manipulator. Interestingly, results highlight that data efficiency and out of sample accuracy can be significantly improved considering as input space an augmented space built starting from the standard inputs.

The article is organized as follows. In Section II, we describe the setup we consider, and we formally define the problem of interest. In Section III, we first provide some background about robot kinematic, and we derive the three systems of equations which represent the theoretical core of the proposed solution. In Section IV, we describe the strategy we propose for the identification of the kinematic structure. In Section V, based on the identified kinematic structure, we propose a data-driven algorithm for the estimation of the forward kinematics, as well as we design a simple kinematic controller based on the learned model. Finally, in Section VI, we report the obtained experimental results. Section VII concludes this article.

Preliminary results regarding only the kinematic structure identification problem have been reported in [17], where we considered only the noiseless measurements case; moreover, in [17], we did not prove the convergence of the classification algorithm.

II. Setup Description and Problem Formulation

In this section, we formally describe the setup and the problem we aim at solving. The framework is the same one adopted in [13] which consists of a camera and a robotic arm with n links and $n-1$ joints, forming an open kinematic chain. Let $\boldsymbol{q}(t) \in \mathbb{R}^{n-1}$ be the vector of joint positions at time t, measured through the motor encoders; the kth component of $\boldsymbol{q}(t)$, $k = 1, \ldots, n-1$, is denoted by $q_k(t)$. In this work, we consider only two types of joint, revolute and prismatic. More complex connections are rarely adopted, and in case they can be derived combining these two types. Concerning the camera placement, we adopt an eyes-to-hand configuration, i.e., the camera is fixed and it is observing the robot; the camera reference frame (RF) coincides with the world RF. It is worth mentioning that assumptions on the camera placement can be relaxed. In particular, as discussed in Section IV-A and V, the proposed strategy can be applied also when the camera is moving, and even if its pose is not known. However, for simplicity, we will consider the camera fixed.

The poses of the links are measured using fiducial markers, see for instance [18]–[20]. More precisely, a distinct fiducial marker M_i, $i = 1, \ldots, n$, is attached to each of the L_j, $j = 1, \ldots, n$, links, where the subscript j denotes the position of the link in the kinematic chain (L_1 and L_n are, respectively, the first and the last link). A pictorial representation of the overall setup is depicted in Fig. 1, while, in Fig. 2, we have reported a frame collected with the camera. It should be noted that we have used different subscripts for links, markers, and joints. This is to stress that we do not know *a priori* which is the sequence of joints and markers related to the kinematic chain. For example, given link L_j, we do not know the index i of the corresponding marker M_i, as well as the index k of the joint connecting L_j and L_{j+1}.

Measurements of the markers' poses in the camera RF are obtained applying solutions described in [18], [19], and [20], processing frames acquired by the camera. Let $\boldsymbol{p}^c_{M_i}(t)$ be the pose of marker M_i at time t, denoted w.r.t. the camera RF, hereafter indicated by c. Then we have

$$\boldsymbol{p}^c_{M_i}(t) = \begin{bmatrix} \boldsymbol{l}^c_{M_i}(t)^T & \boldsymbol{o}^c_{M_i}(t)^T \end{bmatrix}^T$$

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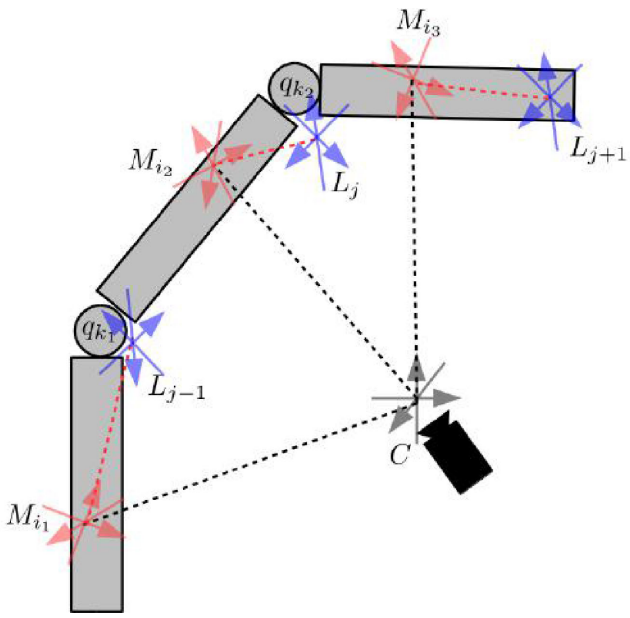

Fig. 1. Symbolic representation of the setup. C, L, M, and q point out, respectively, the camera, the link, and the markers RF and the joint signals.

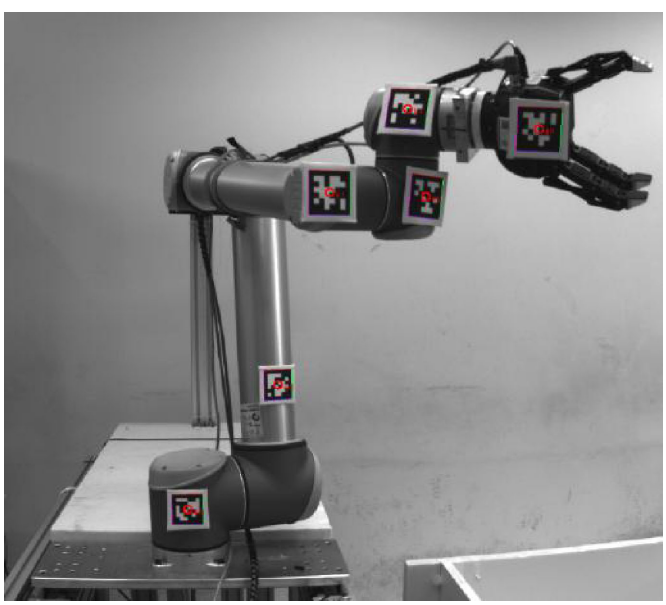

Fig. 2. Frame collected from the camera with the robot and the fiducial markers visible.

where $\boldsymbol{l}^c_{M_i}$ is the vector collecting the three Cartesian coordinates $x^c_{M_i}$, $y^c_{M_i}$, and $z^c_{M_i}$ of the relative translation, while $\boldsymbol{o}^c_{M_i}$ expresses the relative orientation in yaw-pitch-roll convention; namely, $\boldsymbol{o}^c_{M_i} = [\gamma^c_{M_i}(t),\ \beta^c_{M_i}(t),\ \alpha^c_{M_i}(t)]$, where $\gamma^c_{M_i}(t)$, $\beta^c_{M_i}(t)$, and $\alpha^c_{M_i}(t)$, denote, respectively, the yaw, pitch, and roll angles. For future use, we introduce also an alternative expression of relative orientations, based on rotation matrices. Specifically, consider marker M_i, then its orientation in the camera RF can be described by the rotation matrix $R^c_{M_i}$, where

$$R^c_{M_i} = R_z(\gamma^c_{M_i})R_y(\beta^c_{M_i})R_x(\alpha^c_{M_i})$$

being R_z, R_y, and R_x the elementary rotation matrices around the z, y, and x axes, respectively. Now let $P(t)$ be the collection of marker poses at time t, that is

$$P(t) = \{\boldsymbol{p}^c_{M_1}(t), \ldots, \boldsymbol{p}^c_{M_n}(t)\}.$$

Then, given T observations taken at time instants $t_1, \ldots, t_T$, the overall measurements available are

$$\mathcal{D} = \{(P(t_1), \boldsymbol{q}(t_1)), \ldots, (P(t_T), \boldsymbol{q}(t_T))\}. \tag{1}$$

For future use, we introduce also the following sets:

$$P_i = \{\boldsymbol{p}^c_{M_i}(t), t = t_1, \ldots, t_T\}$$
$$\boldsymbol{q}_k = \{q_k(t), t = t_1, \ldots, t_T\}.$$

From now on, to keep the notation compact, we make explicit the dependency on time only when necessary.

The main goal of this work is deriving a forward kinematics of the manipulator, i.e., a model describing the pose of the marker attached to the last link as a function of the joints variables. The solution we propose consists in the following two steps. First, starting from $\mathcal{D}$, we apply a procedure able to identify the kinematic structure of the manipulator (see Sections III and IV). More formally, the kinematic structure identification problem consists in a classification problem, decomposable into three subtasks.

1) Identifying $\mathcal{S}_M = \{M_{i_1}, \ldots, M_{i_n}\}$, i.e., the sequence of markers associated to the kinematic chain $L_1, \ldots, L_n$.
2) Identifying $\mathcal{S}_Q = \{Q_{k_1}, \ldots, Q_{k_{n-1}}\}$, i.e., the sequence of joint types connecting consecutive links along the kinematic chain, starting from the couple (L_1, L_2), up to the couple (L_{n-1}, L_n); more precisely, Q_{k_j} is a binary variable assuming value 0 (resp. 1) when the joint between L_j and L_{j+1} is prismatic (resp. revolute).
3) Identifying $\mathcal{S}_q = \{q_{k_1}, \ldots, q_{k_{n-1}}\}$, i.e., the sequence of joint signals.

In the second step, relying on the knowledge of $\mathcal{S}_Q$, we run a data-driven algorithm based on GPR to derive a model of the forward kinematics (see Section V).

III. Relations Between Subsequent Markers

In this section, we derive the theoretical building blocks of our kinematic structure identification algorithm. We consider the ideal condition of knowing the real values of $\boldsymbol{q}$ and $\boldsymbol{p}^c_{M_i}$. We start considering the case that markers M_{i_1} and M_{i_2} are attached to the consecutive links L_{j_1}, L_{j_2}, connected through joint q_k. Without loss of generality, we can assume $j_2 = j_1 + 1$. Elaborating standard kinematic relations involved by such assumption, and distinguishing the two cases, q_k is prismatic or revolute, we derive three systems of equations that are verified when the triplet (M_{i_1}, M_{i_2}, q_k) is connected. Before going into the details, we briefly review some useful background notions about Kinematics.

A. Background

To provide a mathematical description of the transformations occurring along the kinematic chain we need to define an RF for

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each link and for each maker. As far as the links are concerned, we adopt the DH convention; for details we refer the interested reader to [1, Ch. 2.8.2], chapter 2.8.2. Once the RFs of the links have been assigned, the expression of $R_{L_{j-1}}^{L_j}$, i.e., the relative orientation between the consecutive links L_{j-1} and L_j, is given by

$$R_{L_j}^{L_{j-1}} = R_z(\theta_j)R_x(\alpha_j) \tag{2}$$

where α_j is a constant parameter (see [1, Fig. 2.16, p. 62] for a pictorial description of α_j and θ_j). In case the joint connecting L_{j-1} and L_j is prismatic, then θ_j is constant and equal to θ_j^0, while, if the joint is revolute and controlled by q_k, it holds $\theta_j = \theta_j^0 + q_k$. The relation between the relative positions of L_{j-1} and L_j is described by $\boldsymbol{l}_{L_j}^{L_{j-1}}$, i.e., the expression of the origin of L_j-RF w.r.t. the origin of L_{j-1}-RF. We have that

$$\boldsymbol{l}_{L_j}^{L_{j-1}} = \begin{bmatrix} 0 \\ 0 \\ d_j \end{bmatrix} + R_z(\theta_j) \begin{bmatrix} a_j \\ 0 \\ 0 \end{bmatrix} \tag{3}$$

where $R_z(\theta_j)$ is defined as before, and a_j is a constant parameter of the kinematic (see [1, Fig. 2.16, p. 62] for a pictorial description of a_j and d_j). If the joint connecting L_{j-1} and L_j is revolute then d_j is constant and equal to d_j^0, while, if it is prismatic and parametrized by q_k, then it holds $d_j = d_j^0 + q_k$.

Additionally, we need to describe the pose of each marker w.r.t. the RF of the link it is attached to. For example, if M_{i_1} is attached to L_{j_1}, then the position and the orientation of M_{i_1} w.r.t. L_{j_1} are described, respectively, by $\boldsymbol{l}_{M_{i_1}}^{L_{j_1}}$ and $R_{M_{i_1}}^{L_{j_1}}$. Observe that, since the markers are assumed to be rigidly attached to the corresponding links, $\boldsymbol{l}_{M_{i_1}}^{L_{j_1}}$ and $R_{M_{i_1}}^{L_{j_1}}$ are constant and independent of the joint signals. For later use, it is convenient to introduce also $R_{L_{j_1}}^{M_{i_1}} = (R_{M_{i_1}}^{L_{j_1}})^T$ and $\boldsymbol{l}_{L_{j_1}}^{M_{i_1}}$. Similar definitions hold for M_{i_2} and L_{j_2}, assuming M_{i_2} is attached to L_{j_2}.

We stress that $\boldsymbol{l}_{M_{i_1}}^{L_{j_1}}$ and $R_{M_{i_1}}^{L_{j_1}}$ are unknown, and that we do not introduce any limitation on the way the markers are attached to the links. From a practical point of view, this fact is very interesting, since it allows adopting the proposed algorithm even in setups different from the one we described in Section II. For instance, it might happen that fiducial markers are not available and the use of *ad hoc* computer-vision algorithms is required to detect and track fictitious markers, that is, other features that identify the robot links, [21]–[23]. In this context, the markers placement is not controllable, but it still holds that $\boldsymbol{l}_{M_{i_1}}^{L_{j_1}}$ and $R_{M_{i_1}}^{L_{j_1}}$ are constant.

Based on the quantities introduced above, and exploiting standard algebraic properties of rotation matrices, we derive two relations describing the relative pose between consecutive markers. Let $\boldsymbol{l}_{M_{i_2}}^{M_{i_1}}$ and $R_{M_{i_2}}^{M_{i_1}}$ be, respectively, the Cartesian coordinates of the origin of M_{i_2}-RF w.r.t. M_{i_1}-RF, and the relative orientation between M_{i_2}-RF and M_{i_1}-RF. Then the following relations hold:

$$\boldsymbol{l}_{M_{i_2}}^{M_{i_1}} = \boldsymbol{l}_{L_{j_1}}^{M_{i_1}} + R_{L_{j_1}}^{M_{i_1}} \boldsymbol{l}_{L_{j_2}}^{L_{j_1}} + R_{M_{i_2}}^{M_{i_1}} \left(-\boldsymbol{l}_{L_{j_2}}^{M_{i_2}}\right) \tag{4}$$

and

$$R_{M_{i_2}}^{M_{i_1}} = R_{L_{j_1}}^{M_{i_1}} R_{L_{j_2}}^{L_{j_1}}(\theta_{j_2}) R_{M_{i_2}}^{L_{j_2}}. \tag{5}$$

Notice that quantities $\boldsymbol{l}_{M_{i_2}}^{M_{i_1}}$ and $R_{M_{i_2}}^{M_{i_1}}$ are indirectly known, since

$$\boldsymbol{l}_{M_{i_2}}^{M_{i_1}} = (R_{M_{i_1}}^c)^T (\boldsymbol{l}_{M_{i_2}}^c - \boldsymbol{l}_{M_{i_1}}^c)$$
$$R_{M_{i_2}}^{M_{i_1}} = (R_{M_{i_1}}^c)^T R_{M_{i_2}}^c$$

where $R_{M_{i_1}}^c$, $R_{M_{i_2}}^c$, $\boldsymbol{l}_{M_{i_1}}^c$, and $\boldsymbol{l}_{M_{i_2}}^c$ are obtained processing the information coming from the camera, as outlined in Section II. In the following the mathematical analysis of (4) and (5) assuming a prismatic joint or a revolute joint connecting two consecutive links will be detailed.

B. Prismatic Joint

Assume that the joint connecting L_{j_1} and L_{j_2} is prismatic and let q_k be the corresponding joint variable. In this case the angle θ_j is constant, thus it follows that the relative orientation between L_{j_1} and L_{j_2} is not affected by variations of the joint variable, that is, the matrix $R_{M_{i_2}}^{M_{i_1}}$ is also constant over time. By substituting the expression of $\boldsymbol{l}_{L_{j_2}}^{L_{j_1}}$ given in (3) into (4), the following equation holds:

$$\begin{aligned}
\boldsymbol{l}_{M_{i_2}}^{M_{i_1}} &= \boldsymbol{l}_{L_{j_1}}^{M_{i_1}} + R_{L_{j_1}}^{M_{i_1}} \left(\begin{bmatrix} 0 \\ 0 \\ d_j^0 + q_k \end{bmatrix} + R_z(\theta_j^0) \begin{bmatrix} a_j^0 \\ 0 \\ 0 \end{bmatrix} \right) \\
&\quad + R_{M_{i_2}}^{M_{i_1}} \left(-\boldsymbol{l}_{L_{j_2}}^{M_{i_2}}\right) \\
&= \boldsymbol{l}_{L_{j_1}}^{M_{i_1}} + R_{L_{j_1}}^{M_{i_1}} \left(\begin{bmatrix} 0 \\ 0 \\ d_j^0 \end{bmatrix} + R_z(\theta_j^0) \begin{bmatrix} a_j^0 \\ 0 \\ 0 \end{bmatrix} \right) \\
&\quad + R_{M_{i_2}}^{M_{i_1}} \left(-\boldsymbol{l}_{L_{j_2}}^{M_{i_2}}\right) + R_{L_{j_1}}^{M_{i_1}} \begin{bmatrix} 0 \\ 0 \\ q_k \end{bmatrix}
\end{aligned} \tag{6}$$

where the first three addenda of the right-hand side of the last equation are constant, and they can be compacted in the vector $\boldsymbol{l}_{i_2}^{i_1}$, while the last term has constant direction and module depending on the joint coordinate q_k. In addition, observe that (6) is linear w.r.t. $\boldsymbol{l}_{i_2}^{i_1}$ and the third column of $R_{L_{j_1}}^{M_{i_1}}$, that we denote hereafter by $\boldsymbol{z}_{L_{j_1}}^{M_{i_1}}$. Then, we can write

$$\boldsymbol{l}_{M_{i_2}}^{M_{i_1}} = \begin{bmatrix} I_3 & q_k I_3 \end{bmatrix} \begin{bmatrix} \boldsymbol{l}_{i_2}^{i_1} \\ \left(\boldsymbol{z}_{L_{j_1}}^{M_{i_1}}\right) \end{bmatrix} = A(q_k)\boldsymbol{b}_{i_1}^{i_2} \tag{7}$$

where

$$A(q_k) = \begin{bmatrix} I_3 & q_k I_3 \end{bmatrix}$$

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is the vector of regressors, and

$$\boldsymbol{b}_{i_1}^{i_2} = \begin{bmatrix} \boldsymbol{l}_{i_2}^{i_1} \\ \\ \boldsymbol{z}_{L_{j_1}}^{M_{i_1}} \end{bmatrix}$$

is the vector collecting the parameters of the linear system. Since the last three elements of $\boldsymbol{b}_{i_1}^{i_2}$ are the columns of a rotation matrix, we have that the following constraint must be satisfied:

$$\left(\boldsymbol{z}_{L_{j_1}}^{M_{i_1}}\right)^T \boldsymbol{z}_{L_{j_1}}^{M_{i_1}} = 1.$$

It is easy to see that, by introducing the block diagonal matrix $H = \mathrm{blockdiag}(0_{3\times 3}, I_3)$, where $0_{3\times 3}$ is a 3×3 matrix of zeros, the last equation can be written in terms of $\boldsymbol{b}_{i_1}^{i_2}$, as

$$\left(\boldsymbol{b}_{i_1}^{i_2}\right)^T H \boldsymbol{b}_{i_1}^{i_2} = 1. \tag{8}$$

If, instead of considering a single observation, we consider the T observations in the set $\mathcal{D}$ defined in (1), we obtain the following linear system of $3T$ equations

$$\boldsymbol{l}\left(P_{i_1}, P_{i_2}\right) = \begin{bmatrix} A\left(q_k(t_1)\right) \\ \vdots \\ A\left(q_k(t_T)\right) \end{bmatrix} \boldsymbol{b}_{i_1}^{i_2} = \mathcal{A}(\boldsymbol{q}_k)\boldsymbol{b}_{i_1}^{i_2} \tag{9}$$

where $\boldsymbol{l}(P_{i_1}, P_{i_2})$ is the vector collecting the relative translations between markers M_{i_1} and M_{i_2}, that is

$$\boldsymbol{l}\left(P_{i_1}, P_{i_2}\right) = \left[(\boldsymbol{l}_{M_{i_2}}^{M_{i_1}}(t_1))^T, \ldots, (\boldsymbol{l}_{M_{i_2}}^{M_{i_1}}(t_T))^T\right]^T.$$

We remark that $\boldsymbol{l}(P_{i_1}, P_{i_2})$ contains information that can be indirectly measured, $\mathcal{A}(\boldsymbol{q}_k)$ is known, while $\boldsymbol{b}_{i_1}^{i_2}$ is the vector of the unknown variables. We have the following Proposition.

Proposition 1: Consider two markers M_{i_1} and M_{i_2}, attached to consecutive links connected through a prismatic joint, whose corresponding joint signal is q_k. Then, given a set of observations $\mathcal{D}$, the relative orientation between markers is constant, i.e., $R_{M_{i_2}}^{M_{i_1}}$ is constant, and the linear system in (9) admits solution in $\boldsymbol{b}_{i_1}^{i_2}$ satisfying the constraint in (8).

C. Revolute Joint

Now assume that the joint connecting L_{j_1} and L_{j_2} is revolute and that q_k is the corresponding joint variable. Starting from (4), and substituting $\boldsymbol{l}_{L_{j_2}}^{L_{j_1}}$ with the expression in (3), we obtain

$$\begin{aligned} \boldsymbol{l}_{M_{i_2}}^{M_{i_1}} &= \boldsymbol{l}_{L_{j_1}}^{M_{i_1}} + R_{L_{j_1}}^{M_{i_1}} \left(\begin{bmatrix} 0 \\ 0 \\ d_{j_2} \end{bmatrix} + R_{L_{j_2}}^{L_{j_1}}(\theta_{j_2}) \begin{bmatrix} a_{j_2} \\ 0 \\ 0 \end{bmatrix} \right) \\ &\quad + R_{M_{i_2}}^{M_{i_1}} \left(-\boldsymbol{l}_{L_{j_2}}^{M_{i_2}}\right) \\ &= \boldsymbol{l}_{L_{j_1}}^{M_{i_1}} + R_{L_{j_1}}^{M_{i_1}} \begin{bmatrix} 0 \\ 0 \\ d_{j_2} \end{bmatrix} + R_{M_{i_2}}^{M_{i_1}} R_{L_{j_2}}^{M_{i_2}} \begin{bmatrix} a_{j_2} \\ 0 \\ 0 \end{bmatrix} \\ &\quad + R_{M_{i_2}}^{M_{i_1}} \left(-\boldsymbol{l}_{L_{j_2}}^{M_{i_2}}\right) \end{aligned} \tag{10}$$

where, as before, the different contributions have been grouped conveniently. In particular the first two terms are constant, while the last two are linear w.r.t. $R_{M_{j_2}}^{M_{j_1}}$. As a consequence, equation (10) is linear w.r.t. the vector of variables

$$\bar{\boldsymbol{b}}_{i_2}^{i_1} = \begin{bmatrix} \boldsymbol{l}_{L_{j_1}}^{M_{i_1}} + R_{L_{j_1}}^{M_{i_1}} \begin{bmatrix} 0 \\ 0 \\ d_{j_2} \end{bmatrix} \\ \\ R_{L_{j_2}}^{M_{i_1}} \begin{bmatrix} a_{j_2} \\ 0 \\ 0 \end{bmatrix} - \boldsymbol{l}_{L_{j_2}}^{M_{i_2}} \end{bmatrix}$$

given that (10) can be rewritten as

$$\begin{aligned} \boldsymbol{l}_{M_{i_2}}^{M_{i_1}} &= \begin{bmatrix} I_3 & R_{M_{i_2}}^{M_{i_1}} \end{bmatrix} \begin{bmatrix} \boldsymbol{l}_{L_{j_1}}^{M_{i_1}} + R_{L_{j_1}}^{M_{i_1}} \begin{bmatrix} 0 \\ 0 \\ d_{j_2} \end{bmatrix} \\ \\ R_{L_{j_2}}^{M_{i_1}} \begin{bmatrix} a_{j_2} \\ 0 \\ 0 \end{bmatrix} - \boldsymbol{l}_{L_{j_2}}^{M_{i_2}} \end{bmatrix} \\ &= \bar{A}(\boldsymbol{p}_{M_{i_1}}^c, \boldsymbol{p}_{M_{i_2}}^c)\bar{\boldsymbol{b}}_{i_2}^{i_1}. \end{aligned} \tag{11}$$

The last equation decomposes $\boldsymbol{l}_{M_{i_2}}^{M_{i_1}}$ into the sum of two vectors, one that is constant, and one with constant module and direction depending on $R_{L_{j_1}}^{M_{i_1}}$.

When considering the set of observations $\mathcal{D}$, we obtain the following $3T$ linear equations

$$\begin{aligned} \boldsymbol{l}\left(P_{i_1}, P_{i_2}\right) &= \begin{bmatrix} \bar{A}\left(\boldsymbol{p}_{M_{i_1}}^c(t_1), \boldsymbol{p}_{M_{i_2}}^c(t_1)\right) \\ \vdots \\ \bar{A}\left(\boldsymbol{p}_{M_{i_1}}^c(t_T), \boldsymbol{p}_{M_{i_2}}^c(t_T)\right) \end{bmatrix} \bar{\boldsymbol{b}}_{i_2}^{i_1} \\ &= \bar{\mathcal{A}}(P_{i_1}, P_{i_2})\bar{\boldsymbol{b}}_{i_2}^{i_1} \end{aligned} \tag{12}$$

where $\boldsymbol{l}(P_{i_1}, P_{i_2})$ and $\bar{\mathcal{A}}(P_{i_1}, P_{i_2})$ are known, while $\bar{\boldsymbol{b}}_{i_2}^{i_1}$ is the vector of unknown variables. We have the following Proposition.

Proposition 2: Let M_{i_1} and M_{i_2} be two markers attached to consecutive links, and assume that the joint between the two links is revolute. Then, the linear system of equations defined by the set of observations $\mathcal{D}$ in (12) admits solution in $\bar{\boldsymbol{b}}_{i_2}^{i_1}$.

It is worth stressing that (12) defines a set of conditions related to the relative translation between markers. Concerning the relative orientation, we observe that, differently from the prismatic case, the matrix $R_{L_{j_2}}^{L_{j_1}}$ is not constant, since it depends explicitly on the joint signal q_k. In particular, expanding $R_{L_{j_2}}^{L_{j_1}}$ using the expression in (2), we can rewrite (5), i.e., the relative orientation between markers, as

$$R_{M_{i_2}}^{M_{i_1}} = R_{L_{j_1}}^{M_{i_1}} R_z(q_k + \theta_{j_2}^0) R_x(\alpha_{j_2}) R_{M_{i_2}}^{L_{j_2}} = R_{L_{j_1}}^{M_{i_1}} R_z(q_k) R_{M_{i_2}}^{\bar{L}_{j_2}}$$

where, exploiting the properties of elementary rotation matrices, we have $R_{M_{i_2}}^{\bar{L}_{j_2}} = R_z(\theta_{j_2}^0) R_x(\alpha_{j_2}) R_{M_{i_2}}^{L_{j_2}}$.

The previous matrix equation defines nine nonlinear equations in the elements of the unknown matrices $R_{M_{i_2}}^{\bar{L}_{j_2}}$ and $R_{L_{j_1}}^{M_{i_1}}$.

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Additionally, since $R^{\bar{L}_{j_2}}_{M_{i_2}}$ and $R^{M_{i_1}}_{L_{j_1}}$ are rotation matrices, the following constraints must be satisfied:

$$R^{M_{i_1}}_{L_{j_1}}\left(R^{M_{i_1}}_{L_{j_1}}\right)^T = I_3\,,\; R^{\bar{L}_{j_2}}_{M_{i_2}}\left(R^{\bar{L}_{j_2}}_{M_{i_2}}\right)^T = I_3.$$

A more compact description can be obtained relying on the use of rotation angles. First, instead of considering as unknowns the elements of $R^{\bar{L}_{j_2}}_{M_{i_2}}$ and $R^{M_{i_1}}_{L_{j_1}}$, we describe both rotation matrices using the yaw, pitch, and roll angles, collected in the two vectors

$$\boldsymbol{o}^{M_{i_1}}_{L_{j_1}} = \left[\gamma^{M_{i_1}}_{L_{j_1}}, \beta^{M_{i_1}}_{L_{j_1}}, \alpha^{M_{i_1}}_{L_{j_1}}\right]^T, \boldsymbol{o}^{\bar{L}_{j_2}}_{M_{i_2}} = \left[\gamma^{\bar{L}_{j_2}}_{M_{i_2}}, \beta^{\bar{L}_{j_2}}_{M_{i_2}}, \alpha^{\bar{L}_{j_2}}_{M_{i_2}}\right]^T.$$

Then we have

$$\begin{aligned} R^{M_{i_1}}_{L_{j_1}} &= R_z(\gamma^{M_{i_1}}_{L_{j_1}})R_y(\beta^{M_{i_1}}_{L_{j_1}})R_x(\alpha^{M_{i_1}}_{L_{j_1}}) \\ R^{\bar{L}_{j_2}}_{M_{i_2}} &= R_z(\gamma^{\bar{L}_{j_2}}_{M_{i_2}})R_y(\beta^{\bar{L}_{j_2}}_{M_{i_2}})R_x(\alpha^{\bar{L}_{j_2}}_{M_{i_2}}). \end{aligned} \tag{13}$$

In this way, we enforce orthogonality, while reducing the number of unknowns of the system. Second, also (Section III-C) can be rewritten w.r.t. the yaw, pitch, and roll angles, obtaining

$$\boldsymbol{o}^{M_{i_1}}_{M_{i_2}} = f_o\left(R^{M_{i_1}}_{L_{j_1}} R_z(q_k) R^{\bar{L}_{j_2}}_{M_{i_2}}\right)$$

where f_o is a function that converts a rotation matrix in yaw, pitch, and roll angles.

If, instead of considering a single observation, we consider a set of observations $\mathcal{D}$, we obtain $3T$ nonlinear equations

$$\begin{bmatrix} \boldsymbol{o}^{M_{i_1}}_{M_{i_2}}(t_1) \\ \vdots \\ \boldsymbol{o}^{M_{i_1}}_{M_{i_2}}(t_T) \end{bmatrix} = \begin{bmatrix} f_o\left(R^{M_{i_1}}_{L_{j_1}} R_z\left(q_k(t_1)\right) R^{\bar{L}_{j_2}}_{M_{i_2}}\right) \\ \vdots \\ f_o\left(R^{M_{i_1}}_{L_{j_1}} R_z\left(q_k(t_T)\right) R^{\bar{L}_{j_2}}_{M_{i_2}}\right) \end{bmatrix}.$$

In the following, for compactness, we denote the left- and the right-hand side of the previous equation by, respectively, $\boldsymbol{o}(P_{i_1}, P_{i_2})$ and $f_o(\boldsymbol{q}_k, R^{M_{i_1}}_{L_{j_1}}, R^{\bar{L}_{j_2}}_{M_{i_2}})$, obtaining

$$\boldsymbol{o}\left(P_{i_1}, P_{i_2}\right) = f_o(\boldsymbol{q}_k, R^{M_{i_1}}_{L_{j_1}}, R^{\bar{L}_{j_2}}_{M_{i_2}}). \tag{14}$$

We have the following proposition.

Proposition 3: Let M_{i_1} and M_{i_2} be two markers attached to consecutive links connected through the revolute joint q_k. Then, given a set of observations $\mathcal{D}$, the corresponding nonlinear system of equations defined by (13) and (14) admits a solution, i.e., there exist two triplets of yaw, pitch, and roll angles defining $R^{M_{i_1}}_{L_{j_1}}$ and $R^{\bar{L}_{j_2}}_{M_{i_2}}$ satisfying (14).

D. Fully Informative Trajectories

In the previous part of this Section, we have stated three propositions defining conditions that are verified when M_{i_1} and M_{i_2} are attached to consecutive links. In general, the reverse relations are not true since it is possible to exhibit sets of observations $\mathcal{D}$, such that the conditions of the previous propositions are satisfied even if the markers are not attached to subsequent links. In the following we will refer to such cases as *false positives*.

False positive observations are strictly related to the sequences of joint configurations which have generated the set of observations $\mathcal{D}$. In the following, we denote by *fully informative* observations a set of observations $\mathcal{D}$, such that conditions defined in Proposition 1, for the prismatic joint, and in Propositions 2 and 3, for the revolute joint, are necessary and sufficient to verify if two markers are attached to consecutive links. A full characterization of this class is not trivial, however, in this article, we provide a class of input trajectories from which it is possible to properly select a set of *fully informative* observations.

Definition 1: Consider a collection of $n-1$ joints' trajectories, where in each trajectory only one joint is actuated while all the others are kept stuck. For $k = 1, \ldots, n-1$, assume that the kth trajectory is obtained varying the joint signal q_k, and that there exist at least two time instants $t_{1,k}$ and $t_{2,k}$, such that

$$mod\left(q_k(t_{1,k})\right) \neq \mathrm{mod}\left(q_k(t_{2,k})\right) \tag{15}$$

and $q_w(t_{1,k}) = q_w(t_{2,k}), w \neq k$, where $\mathrm{mod}(\cdot)$ is the 2π module operator. Then, we define the set $\bar{\mathcal{D}}$, as the collection of the pairs $(P(t_{1,k}), \boldsymbol{q}(t_{1,k})), (P(t_{2,k}), \boldsymbol{q}(t_{2,k}))$, $k = 1, \ldots, n-1$, that is

$$\bar{\mathcal{D}} = \left\{(P(t_{1,k}), \boldsymbol{q}(t_{1,k})), (P(t_{2,k}), \boldsymbol{q}(t_{2,k}))\right\}_{k=1,\ldots,n-1}.$$

The set $\bar{\mathcal{D}}$ is composed of $n-1$ pair of observations, where for each observation only a single joint is actuated. Condition defined in (15) ensures that for each pair, the two joint configurations are different. Then, $\bar{\mathcal{D}}$ has $2(n-1)$ observations, which for typical robots (i.e., $n = 3, 4, 5, 6, 7$) represents a limited number of observations. It is possible to show that $\bar{\mathcal{D}}$ is a fully informative set, and, in particular, we have the following results.

Proposition 4: Let (M_{i_1}, M_{i_2}, q_k) be a triplet satisfying conditions of Proposition 1, for the set of observations $\bar{\mathcal{D}}$ defined in Definition 1. Then, the corresponding links L_{j_1} and L_{j_2} are subsequent in the kinematic chain, and the joint connecting them is prismatic with input signal q_k.

Proposition 5: Let (M_{i_1}, M_{i_2}, q_k) be a triplet satisfying equations in Propositions 2 and 3, for the set of observations $\bar{\mathcal{D}}$ defined in Definition 1. Then, the corresponding links L_{j_1} and L_{j_2} are subsequent in the kinematic chain, and the joint connecting them is revolute with input signal q_k.

The proofs of the above Propositions are reported in the Appendix. It is important to observe that Propositions 4 and 5 require suitable excitation assumptions, that might not be verified for general input trajectories. As a consequence, in order to apply the results of the two propositions, an ad-hoc actuation strategy should be designed to collect observations satisfying conditions in Definition 1. However, we would like to point out that the set of observations introduced in Definition 1 represents just a small subset of the class of *fully informative* observations. Indeed, through extensive Monte Carlo simulations, reported in Section VI-A, we show that selecting a set of observations not *fully informative* from generic trajectories seems to be a very unlikely event. In order not to interrupt the flow of the presentation, we leave this discussion to Section VI-A. In the next section, we describe our kinematic structure classification algorithm, assuming that the observations collected are *fully informative*, regardless of whether trajectories belong to the class in Definition 1 or not.

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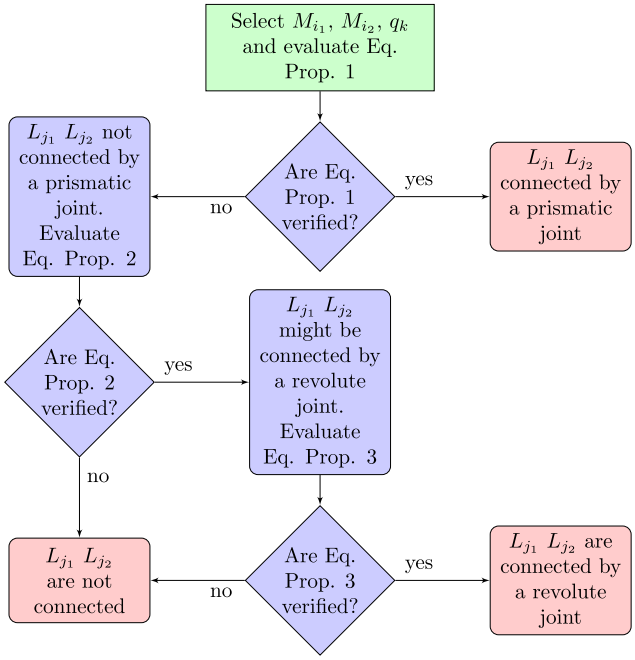

Fig. 3. Flow chart of an iteration of the proposed algorithm. Ending conditions and the starting block are highlighted, respectively, in red and green.

IV. Proposed Approach for Kinematic Structure Identification

In this section, we describe the algorithm we propose to deal with the robot kinematic structure classification problem. First, driven by Propositions 4 and 5, we propose an iterative algorithm for the ideal case, that is, when $\boldsymbol{q}$ and $\boldsymbol{p}^c_{M_i}$ are assumed to be perfectly known (noiseless case). Assuming this ideal condition, as well as that the set of observations $\mathcal{D}$ is *fully informative*, we can prove the convergence of the algorithm to the correct solution. Then, after discussing the sources of noise present in our setup, we modify the algorithm to deal with the nonideal conditions.

A. Kinematic Structure Classification With Ideal Conditions

Results of Propositions 4 and 5 suggest that to determine if a triplet (M_{i_1}, M_{i_2}, q_k) is connected, and in case, if q_k is revolute or prismatic, we can check whether the conditions defined by Propositions 1, 2, and 3 are satisfied. From the computational point of view, a convenient strategy is proposed in Fig. 3. Specifically, given the pair of markers (M_{i_1}, M_{i_2}) and a joint signal q_k, we first evaluate if the markers are connected through a prismatic joint and, in case this test is negative, we evaluate if they are connected through a revolute joint. The first test consists in checking whether the linear equations defined in Proposition 1 are satisfied. The second test, instead, is composed of two steps: The first step verifies if the linear equations of Proposition 2 admit solution. If so, the algorithm proceeds to the second step, solving the system of nonlinear equations of Proposition 3. Observe that, in this way, the last step is performed only if necessary, thus minimizing its executions. This fact is particularly relevant from the computational point of view, since the nonlinear test is the most expensive part of the algorithm.

In order to identify the kinematic structure of the robot, we propose an algorithm that applies iteratively the procedure described in the flow chart in Fig. 3. The algorithm returns $\hat{\mathcal{S}}_M$, $\hat{\mathcal{S}}_Q$, and $\hat{\mathcal{S}}_q$, the estimates of $\mathcal{S}_M$, $\mathcal{S}_Q$, and $\mathcal{S}_q$, namely, the sequences of markers, joint types, and joint signals defining the kinematic structure of the robot. $\hat{\mathcal{S}}_M$, $\hat{\mathcal{S}}_Q$, and $\hat{\mathcal{S}}_q$ are initialized as empty sets. At the beginning, the algorithm identifies the marker associated to the base link, appends it to $\hat{\mathcal{S}}_M$, and set it as the current marker. The identification of this marker is trivial, given that its pose in the camera RF is constant. After that, the algorithm selects a triplet, adding to the current marker a new marker and a joint signal randomly picked among the ones that are not in $\hat{\mathcal{S}}_M$ and $\hat{\mathcal{S}}_q$. Then, the algorithm checks if data associated with the triplet satisfy conditions of Proposition 1, 2, and 3, as described in Fig. 3. Based on the results obtained, $\hat{\mathcal{S}}_M$, $\hat{\mathcal{S}}_Q$, and $\hat{\mathcal{S}}_q$, are updated accordingly: If these conditions are verified, the marker previously selected is set as current marker, and the procedure is iterated over a new triplet.

B. Sources of Noise

The algorithm proposed in the previous subsection can not be directly applied to observations acquired in a real setup, since, due to the presence of noise, the systems of equations defined by Propositions 1, 2, and 3 are, in general, not satisfied even if the triplet considered is connected.

In our setup, the presence of noise is related to several aspects. A significant contribution is introduced by the noise in markers pose estimation. Markers poses are computed processing images coming from the camera, that are corrupted by the sampling noise caused by the image pixels. The effects of such errors in the estimated pose depend on several geometric parameters, like the distance between the camera and the marker and the angles between the camera optical axis and the normals of the markers. The marker detection algorithm also has an influence on the noise of the pose estimation: The use of subpixel accuracy and of temporal information can reduce the noise and modify the noise model [24].

The dependence on several variables, together with the nonlinearity of the sampling error, makes it difficult to derive a rigorous mathematical description of the errors. Several experimental studies [19], [20], [25] confirm the nonlinear behavior of errors, and show that, for a considerable range of distances and inclination angles, markers' poses are estimated very accurately. For instance, in [25], when the marker–camera distance is smaller than 4 m, position errors are in the order of millimeters, while errors on the yaw, pitch, and roll angles are less than 10^{-2} radiants. Experimental evidence shows that, to obtain similar performance when the distance is higher, a certain inclination angle between the marker and the camera is needed, i.e., the angle between the marker normal and the camera axis should be quite high. Finally, results show that outside the angular and distance ranges providing accurate estimates, errors grow rapidly, possibly resulting in the presence of outliers, that can be easily detected.

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We conclude the characterization of noise sources mentioning other two contributions. The first one is the measurement noise of joint positions. Typically, compared to the errors in markers poses, these errors are smaller since encoders measure angular positions with high precision. The second contribution is due to synchronization errors between the camera and the robot. This occurs when a common time reference for camera acquisitions and robot motion is not available. Notice that these errors could be particularly evident when the robot is moving at high velocities, while their effects can be mitigated moving the robot at small velocities.

C. Kinematic Structure Classification With Nonideal Conditions

In this subsection, we modify the algorithm described in Section IV-A to deal with the presence of noise. Our approach is based on the following idea. To exploit the results of Propositions 4 and 5 even with noisy observations, instead of looking for the exact solutions of the systems defined in Propositions 1, 2, and 3, we solve a least squares problem for each of the three systems, minimizing the residuals. Assuming that observations with outliers can be removed, i.e., that the observations' noises are small, it is reasonable to expect a small residual for a system of equations that would be verified in absence of noise, and, on the contrary, a high residual when not verified. Based on this idea, we introduce $f_r(\cdot)$, that is a function depending on the residuals, and a threshold ξ: The test on (M_{i_1}, M_{i_2}, q_k) is assumed to be positive if the value of f_r is lower than ξ, and negative otherwise.

Before formulating the three optimization problems related to Propositions 1, 2, and 3, it is worth providing some more details about f_r and the thresholds. As far as f_r is concerned, we consider the same function for all the three tests. In particular, given a vector of residuals $\boldsymbol{r} \in \mathbb{R}^N$, f_r is defined as

$$f_r(\boldsymbol{r}) = \sqrt{\frac{\boldsymbol{r}^T \boldsymbol{r}}{N}}.$$

The function f_r corresponds to the root of the mean squared residuals, and provides statistics about the magnitude of the residuals. This value is compared with a threshold that characterizes the expected magnitude of the residuals caused by measurement noises. Notice that the outputs in (9) and (12) are distances, while in (14) they are angles. Then, also due to the different type of uncertainties with which translation and orientation are estimated, we introduce two thresholds, ξ_t and ξ_o. The first one provides a bound on the uncertainties about distances, and it is used to test (9) and (12), while the second one refers to the uncertainty about angles, and it is used to test (14). In the remaining part of this section, we describe the optimization problems that the modified algorithm solves for each of the three tests.

1) Prismatic Joint: Given $\boldsymbol{b} \in \mathbb{R}^6$, let $\boldsymbol{r}_{i_1}^{i_2}(\boldsymbol{b})$ be the vector collecting the residuals of linear system in (9), namely

$$\boldsymbol{r}_{i_1}^{i_2}(\boldsymbol{b}) = \boldsymbol{l}(P_{i_1}, P_{i_2}) - \mathcal{A}(\boldsymbol{q}_k)\boldsymbol{b}.$$

Let $\hat{\boldsymbol{b}}_{i_1}^{i_2}$ be a solution of the following constrained linear least squared problem

$$\hat{\boldsymbol{b}}_{i_1}^{i_2} = \arg\min_{\boldsymbol{b} \in \mathcal{H}} \left(\boldsymbol{r}_{i_1}^{i_2}(\boldsymbol{b})\right)^T \boldsymbol{r}_{i_1}^{i_2}(\boldsymbol{b}) \tag{16}$$

where $\mathcal{H} = \{\boldsymbol{b} \in \mathbb{R}^6 \text{ s.t. } \boldsymbol{b}^T H \boldsymbol{b} = 1\}$. Then, the test is assumed to be positive if $f_r(\boldsymbol{r}_{i_1}^{i_2}(\hat{\boldsymbol{b}}_{i_1}^{i_2})) < \xi_t$, and negative otherwise.

Notice that Proposition 4 defines also a condition on the relative orientation between markers, that must be constant. As a consequence, when considering noisy observations, we expect that the elements of $R_{M_{i_2}}^{M_{i_1}}$ have small variance. Based on this observation, in addition to testing the value of f_r, we compute the variance of the measures of $R_{M_{i_2}}^{M_{i_1}}$, and check if the maximum value is below a given threshold.

2) Revolute Joint – Linear Test: Given $\boldsymbol{b} \in \mathbb{R}^6$, let $\bar{\boldsymbol{r}}_{i_1}^{i_2}(\boldsymbol{b})$ be the vector collecting the residuals of the linear system (12), that is

$$\bar{\boldsymbol{r}}_{i_1}^{i_2}(\boldsymbol{b}) = \boldsymbol{l}(P_{i_1}, P_{i_2}) - \bar{\mathcal{A}}(P_{i_1}, P_{i_2})\boldsymbol{b}.$$

Then we solve the following unconstrained linear least squared problem:

$$\hat{\bar{\boldsymbol{b}}}_{i_1}^{i_2} := \arg\min_{\boldsymbol{b}} \left(\bar{\boldsymbol{r}}_{i_1}^{i_2}(\boldsymbol{b})\right)^T \bar{\boldsymbol{r}}_{i_1}^{i_2}(\boldsymbol{b}). \tag{17}$$

Observe that, under the assumption that $\bar{\mathcal{A}}(P_{i_1}, P_{i_2})$ is full rank, $\hat{\bar{\boldsymbol{b}}}_{i_1}^{i_2}$ can be computed in closed form as

$$\hat{\bar{\boldsymbol{b}}}_{i_1}^{i_2} = \left(\bar{\mathcal{A}}^T(P_{i_1}, P_{i_2})\bar{\mathcal{A}}(P_{i_1}, P_{i_2})\right)^{-1} \bar{\mathcal{A}}^T(P_{i_1}, P_{i_2})\, \boldsymbol{l}(P_{i_1}, P_{i_2}).$$

By a close inspection of (12), one can see that $\bar{\mathcal{A}}(P_{i_1}, P_{i_2})$ is full rank, as soon as there are two observations for which the matrix $R_{M_{i_2}}^{M_{i_1}}$ assumes different values; this condition is easily verified in practice.[1] The test is assumed to be positive if $f_r(\hat{\bar{\boldsymbol{b}}}_{i_1}^{i_2}) < \xi_t$, negative otherwise.

3) Revolute Joint – Nonlinear Test: In case the previous test is positive, the algorithm checks the conditions defined by the nonlinear system in (14) to determine if M_{i_1} and M_{i_2} are connected through the revolute joint q_k. Differently from the previous case, the output of the nonlinear system is a vector of angles. To account for periodicity effects, the residuals are defined as

$$\tilde{\boldsymbol{r}}_{i_1}^{i_2}(\boldsymbol{o}_1, \boldsymbol{o}_2) = \operatorname{mod}\left(\boldsymbol{o}\left(P_{i_1}, P_{i_2}\right) - f_o(\boldsymbol{q}_k, \boldsymbol{o}_1, \boldsymbol{o}_2) + \pi\right) - \pi$$

where we recall that $\operatorname{mod}(\cdot)$ is the 2π module operator. With a slight abuse of notation, vectors $\boldsymbol{o}_1$ and $\boldsymbol{o}_2 \in \mathbb{R}^3$ are directly inputted to $f_o(\cdot)$, instead of computing the relative rotation matrices with (13). Then, the algorithm computes $\hat{\boldsymbol{o}}_{L_{j_1}}^{M_{i_1}}$ $\hat{\boldsymbol{o}}_{M_{i_2}}^{L_{j_2}}$ solving the following nonlinear least squares problem:

$$\left(\hat{\boldsymbol{o}}_{L_{j_1}}^{M_{i_1}}, \hat{\boldsymbol{o}}_{M_{i_2}}^{L_{j_2}}\right) = \arg\min_{(\boldsymbol{o}_1, \boldsymbol{o}_2)} \left(\tilde{\boldsymbol{r}}_{i_1}^{i_2}(\boldsymbol{o}_1, \boldsymbol{o}_2)\right)^T \tilde{\boldsymbol{r}}_{i_1}^{i_2}(\boldsymbol{o}_1, \boldsymbol{o}_2). \tag{18}$$

[1]If not verified, we might compute $\hat{\bar{\boldsymbol{b}}}_{i_1}^{i_2}$ resorting to the pseudo inverse of $\bar{\mathcal{A}}^T(P_{i_1}, P_{i_2})\bar{\mathcal{A}}(P_{i_1}, P_{i_2})$.

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The test is assumed to be positive if $f_r(\tilde{\boldsymbol{r}}_{i_1}^{i_2}(\hat{\boldsymbol{o}}_{L_{j_1}}^{M_{i_1}}, \hat{\boldsymbol{o}}_{M_{i_2}}^{L_{j_2}})) < \xi_o$, negative otherwise.

Remark 1: As mentioned in Section II, for simplicity we have assumed that camera is fixed. However, his assumption can be relaxed. In particular, the proposed classification algorithm can be applied also when the camera is moving, and even if its pose is not known. Indeed, to reconstruct the kinematics structure, the proposed solution uses as input data the joint signals and the relative transformations between markers at each time instant, which are independent on the camera placement.

V. Polynomial-Based Approach for Forward Kinematics Identification

In this section, we describe the strategy we propose to identify the forward kinematics. Our solution is a grey-box solution based on GPR; more specifically, it consists in the design of a suitable polynomial kernel. Minimal prior knowledge about the model is required, since, in order to derive the GPR model, we assume to know only the sequence $\mathcal{S}_Q$; no knowledge about the kinematic parameters is required. The section is organized as follows. First, we provide a formal description of the forward kinematics identification problem. Second, we state a proposition that shows how the forward kinematics can be described by a polynomial function defined over a suitable input space. Third, driven by this proposition, we derive a proper GP model. We conclude the section proposing a simple kinematic controller based on the models derived.

A. Forward Kinematics and Polynomial Functions

The forward kinematics identification problem consists in identifying the map that relates the position of the joints with the pose of the last link w.r.t. the camera RF. In the case that the camera RF is not fixed or known, the pose of the last marker can be expressed w.r.t. the RF of the marker attached to the base link, identified with the kinematic structure classification algorithm. In the following, to keep the notation compact, we denote with the subscript M the marker attached to the last link of the robot. Accordingly, the output of the forward kinematics is given by the vector $\boldsymbol{p}_M^c = [(\boldsymbol{l}_M^c)^T, (\boldsymbol{o}_M^c)^T]^T$, or, equivalently, by the pair $(\boldsymbol{l}_{\boldsymbol{M}}^{\boldsymbol{c}}, R_M^c)$, if we express orientations using rotation matrices. For future use, we introduce also the function relating the yaw, pitch, and roll angles collected in $\boldsymbol{o}_M^c$ with the elements of R_M^c

$$\begin{aligned}
\alpha_M^c &= \tan^{-1}\left(R_{M_{(2,1)}}^c / R_{M_{(1,1)}}^c\right)\\
\beta_M^c &= \tan^{-1}\left(-R_{M_{(3,1)}}^c / \sqrt{(R_{M_{(3,2)}}^c)^2 + (R_{M_{(3,3)}}^c)^2}\right)\\
\gamma_M^c &= \tan^{-1}\left(R_{M_{(3,2)}}^c / R_{M_{(3,3)}}^c\right)
\end{aligned} \tag{19}$$

where $R_{M_{(i,j)}}^c$ is the element of R_M^c in the ith row i and jth column (see [1] for the details).

We show that the forward kinematics can be properly described by a polynomial function over an input vector $\boldsymbol{x}$ which depends on $\boldsymbol{q}$. Let N_r and N_p be, respectively, the number of revolute and prismatic joints, and let $\mathcal{I}_r = \{r_1, \ldots, r_{N_r}\}$ and $\mathcal{I}_p = \{p_1, \ldots, p_{N_p}\}$ be the indices of the revolute and prismatic joints. We define the vector $\boldsymbol{x}$ as

$$\boldsymbol{x} = \left[\boldsymbol{q}_c, \boldsymbol{q}_s, \boldsymbol{q}_p\right] \tag{20}$$

where

$$\begin{aligned}
\boldsymbol{q}_c &= \left[\cos(q_{r_1}), \ldots, \cos(q_{r_{N_r}})\right] \in \mathbb{R}^{N_r}\\
\boldsymbol{q}_s &= \left[\sin(q_{r_1}), \ldots, \sin(q_{r_{N_r}})\right] \in \mathbb{R}^{N_r}\\
\boldsymbol{q}_p &= \left[q_{p_1}, \ldots, q_{p_{N_p}}\right] \in \mathbb{R}^{N_p}.
\end{aligned}$$

Moreover, in the following, we denote by q_{c_b}, q_{s_b}, and q_{p_b} the bth component of $\boldsymbol{q}_c$, $\boldsymbol{q}_s$, and $\boldsymbol{q}_p$, respectively, and by X the set collecting all the $\boldsymbol{x}(t_1), \ldots, \boldsymbol{x}(t_T)$. We have the following result.

Proposition 6: Consider a manipulator with N_r revolute joints and N_p prismatic joints, and let $\mathcal{I}_r$ and $\mathcal{I}_p$ be, respectively, the indices of the revolute and prismatic joints. Then, the elements of vector $\boldsymbol{l}_M^c$ and matrix R_M^c are polynomial functions in the input vector $\boldsymbol{x}$, defined in (20). More precisely,

1) the elements of R_M^c are polynomial functions in the elements of $[\boldsymbol{q}_c, \boldsymbol{q}_s]$, with maximum degree N_r and where each monomial satisfies the conditions

$$\deg(q_{c_b}) + \deg(q_{s_b}) \leq 1, \qquad b = 1, \ldots, N_r;$$

2) the elements of $\boldsymbol{l}_M^c$ are polynomial functions in the elements of $\boldsymbol{x}$ with maximum degree $N_r + N_p$, and where each monomial satisfies the conditions

$$\begin{aligned}
&\deg(q_{p_b}) \leq 1 \qquad b = 1, \ldots, N_p,\\
&\deg(q_{c_b}) + \deg(q_{s_b}) \leq 1 \qquad b = 1, \ldots, N_r.
\end{aligned}$$

The proof is reported in the Appendix.

B. Background on GPR

To achieve robustness w.r.t. noisy observations, we rely on Bayesian techniques, in particular on GPR [26], [27]. In the GPR framework the outputs are modeled as a Gaussian process (GP). Let $\boldsymbol{x}(t)$ and $y(t)$ be, respectively, the input and noisy outputs of function f at time t. Then we have

$$\boldsymbol{y} = \begin{bmatrix} y(t_1)\\ \vdots \\ y(t_T)\end{bmatrix} = \begin{bmatrix} f(\boldsymbol{x}(t_1))\\ \vdots \\ f(\boldsymbol{x}(t_T))\end{bmatrix} + \begin{bmatrix} e(t_1)\\ \vdots \\ e(t_T)\end{bmatrix} = f(X) + \boldsymbol{e}$$

where $e(t_1), \ldots, e(t_T)$ are independent and identically distributed Gaussian noises with zero mean and standard deviation σ, while $f(X) \sim N(0, K(X,X))$, with $K(X,X)$ denoting the kernel matrix, whose elements are defined through the kernel function $k(\cdot,\cdot)$. In particular, the element of the ith row and jth column of K is equal to $k(\boldsymbol{x}(t_i), \boldsymbol{x}(t_j))$. Given a set of N_{tr} training samples $(X, \boldsymbol{y})$, the maximum *a posteriori* probability (MAP) estimator of f is given in closed form. Let $\boldsymbol{x}(t_*)$ be the value of the input location at a generic time instant t_*. Then, the

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MAP estimate of $f(\boldsymbol{x}(t_*))$ is given by

$$\hat{f}(\boldsymbol{q}(t_*)) = \sum_{h=1}^{N_{TR}} \alpha_i k(\boldsymbol{q}(t_*), \boldsymbol{q}(t_h)) \tag{21}$$

where α_i is the ith element of the vector $\boldsymbol{\alpha}$, defined as

$$\boldsymbol{\alpha} = \left(K(X,X) + \sigma^2 I\right)^{-1} \boldsymbol{y}. \tag{22}$$

We conclude this brief review on GPR discussing computational requirements. It is known that the GPR computational cost scales with the cube of the number of training samples; in particular, the bottleneck is given by the computation of the matrix inversion required to compute $\boldsymbol{\alpha}$. Though, this operation needs to be performed only one time off-line, a full GP can be efficiently computed only when the number of samples is in the order of 10^3, or less. When the number of samples is higher, several alternative approaches can be considered. A simple but efficient solution consists in downsampling data. We can apply a constant downsampling, or more efficient solutions, like the one proposed in [28]. Alternatively, we can consider approximation of the GPR model, like the subset of regressors model presented in [29], together with other approximations. Typically, such techniques allows achieving a drastic reduction of the computational cost, without decreasing accuracy of the estimation.

From a functional analysis point of view, Proposition 6 states that the outputs of the forward kinematics are polynomial functions in the augmented space $\boldsymbol{x}$. By designing a suitable kernel function, we can define a regression problem in the reproducing kernel Hilbert space (RKHS), see [27], identified by Proposition 6. We recall two notions that will be used in the next part of this section. The first one is the definition of linear kernel, which is given by the following expression:

$$k(\boldsymbol{x}(t_i), \boldsymbol{x}(t_j)) = \boldsymbol{\phi}^T(\boldsymbol{x}(t_i))\Sigma\boldsymbol{\phi}(\boldsymbol{x}(t_j))$$

where $\boldsymbol{\phi}(\boldsymbol{x}) = [\phi_1(\boldsymbol{x}), \ldots, \phi_p(\boldsymbol{x})] \in \mathbb{R}^p$ is a function mapping $\boldsymbol{x}$ in a given feature space, and $\Sigma \in \mathbb{R}^{p\times p}$ is a positive definite diagonal matrix, whose diagonal elements are the kernel hyperparameters. The basis functions of the RKHS associated to this kernel are the functions in $\boldsymbol{\phi}$. The second notion is that the point-wise product of two kernels is still a kernel. Moreover, consider two linear kernels with feature space

$$\boldsymbol{\phi}^{(1)}(\boldsymbol{x}) = [\phi_1^{(1)}(\boldsymbol{x}), \ldots, \phi_{p_1}^{(1)}(\boldsymbol{x})]$$

$$\boldsymbol{\phi}^{(2)}(\boldsymbol{x}) = [\phi_1^{(2)}(\boldsymbol{x}), \ldots, \phi_{p_2}^{(2)}(\boldsymbol{x})].$$

The basis functions of the RKHS associated to the product of these two linear kernels are all the possible products between an element in $\boldsymbol{\phi}^{(1)}$ and an element in $\boldsymbol{\phi}^{(2)}$ (see [27, Ch. 13.1]), i.e., the vector collecting the following functions:

$$\left\{\phi_i^{(1)}(\boldsymbol{x})\phi_j^{(2)}(\boldsymbol{x}), i = 1, \ldots, p_1, j = 1, \ldots, p_2\right\}.$$

C. Proposed Approach for Forward Kinematics Identification

In our work, we model each element of $\boldsymbol{l}_M^c$ and R_M^c by a distinct GP. For each model we consider different values of the hyperparameters, as well as different types of kernels. Driven by Proposition 6, and considering the point-wise product property discussed above, we define $k_R(\cdot,\cdot)$, that is, the kernel function modeling the elements of R_M^c, as the product of several elementary building blocks. In particular, each building block is a linear kernel in the bth component of $\boldsymbol{q}_c$ and $\boldsymbol{q}_s$, with $b = 1, \ldots, N_r$, accounting for the contributions of $\cos(q_{r_b})$ and $\sin(q_{r_b})$. Formally k_R is defined as

$$k_R(\boldsymbol{x}(t_i), \boldsymbol{x}(t_j)) = \prod_{b=1}^{N_r} \boldsymbol{q}_{cs_b}^T(t_i)\Sigma_{R_b}\boldsymbol{q}_{cs_b}(t_j) \tag{23}$$

where $\boldsymbol{q}_{cs_b} = [1, q_{c_b}, q_{s_b}]$, and $\Sigma_{R_b} \in \mathbb{R}^{3\times 3}$, $b = 1, \ldots, N_r$, are diagonal matrices, whose elements are the hyperparameters of the kernel.[2]

Next, we define $k_l(\cdot,\cdot)$, that is, the kernel function modeling the elements of $\boldsymbol{l}_M^c$. Proposition 6 suggests that, in addition to $\boldsymbol{q}_c, \boldsymbol{q}_s$, we need to account also for the linear contributions of the $\boldsymbol{q}_p$ elements. To do that, we multiply $k_R(\cdot,\cdot)$ by an additional kernel, obtaining

$$k_l\left(\boldsymbol{x}(t_i), \boldsymbol{x}(t_j)\right) = k_R\left(\boldsymbol{x}(t_i), \boldsymbol{x}(t_j)\right)\bar{\boldsymbol{q}}_p^T(t_i)\Sigma_l\bar{\boldsymbol{q}}_p(t_j) \tag{24}$$

where $\bar{\boldsymbol{q}}_p$ is the vector obtained concatenating 1 with $\boldsymbol{q}_p$; the matrix $\Sigma_l \in \mathbb{R}^{(N_p+1)\times(N_p+1)}$ is a diagonal matrix, whose diagonal elements are additional hyperparameters.

We conclude the description of the proposed learning strategy with the following observation. As discussed before, R_M^c is a redundant description of the relative orientation $\boldsymbol{o}_M^c$. To limit the number of models to be learned, in our work we learn only the models of $R_{M_{(1,1)}}^c$, $R_{M_{(2,1)}}^c$, $R_{M_{(3,1)}}^c$, $R_{M_{(3,2)}}^c$, and $R_{M_{(3,3)}}^c$, and we obtain $\boldsymbol{o}_M^c$ applying (19).

D. Kinematic Controller Based on the GPR Model

In this subsection, we introduce a naive kinematic controller based on the forward kinematics model previously described. It is worth stressing that the proposed controller implementation is meant only to test the accuracy of the model obtained, and not to develop a novel control strategy. For this reason we have not addressed several important issues that, instead, should be considered when proposing high performance control strategies.

We assume that each joint motor is equipped with a low level controller able to follow a desired reference trajectory $\boldsymbol{q}_{\text{des}}$, and we denote by δ the sampling time. Typically, when considering trajectories with sufficiently small differences between the desired values at time t and $t+\delta$, the low-level controllers are able to follow accurately the reference trajectory. Our strategy is based on the fact that the model derived in the previous section is differentiable w.r.t. $\boldsymbol{q}$. Then, given $\boldsymbol{p}_*$, a target pose of the last marker, we implement an open loop controller that computes a suitable $\boldsymbol{q}_{\text{des}}$ trajectory driving the robot from the initial pose $\boldsymbol{p}_M^c(0)$ to $\boldsymbol{p}_*$. The proposed controller is defined by the following expression:

$$\boldsymbol{q}_{\text{des}}(t+\delta_T) = \boldsymbol{q}_{\text{des}}(t) - \nu\nabla e_p(t) \tag{25}$$

[2]Notice that the presence of 1 in $\boldsymbol{q}_{cs_b}$ is needed to consider the monomials with degree zero and one in the RKHS defined by $k_R(\cdot,\cdot)$.

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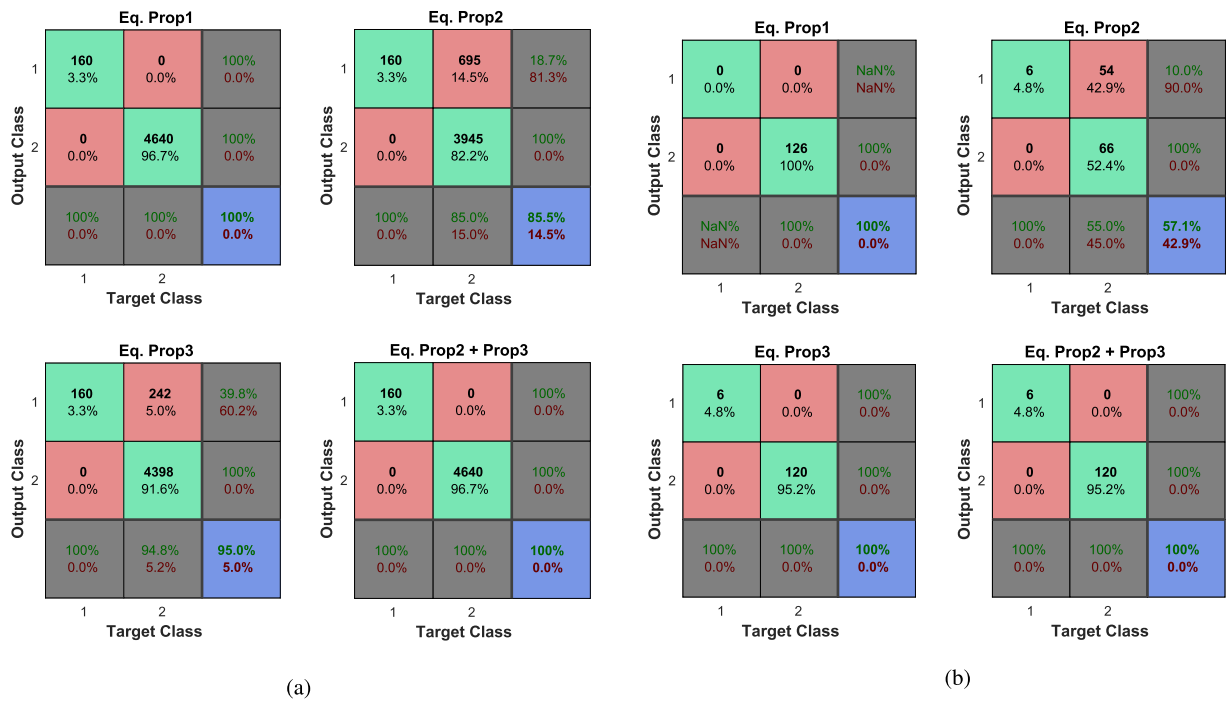

(a) (b)

Fig. 4. Confusion matrices of the experiments in Section VI-A (on the left) and VI-B (on the right). When considering Proposition 1, (resp. Proposition 2 and 3), triplet connected through prismatic (resp. revolute) joint are assigned to class 1. (a) Results of the Monte Carlo simulations in Section VI-A. (b) Results obtained with the UR10 dataset $\mathcal{D}_{tr}$.

where ∇e_p is the gradient, computed w.r.t. to $\boldsymbol{q}$, of the error $e_p(t) = ||\boldsymbol{p}_* - \hat{\boldsymbol{p}}_M^c(t)||$, being $\hat{\boldsymbol{p}}_M^c(t)$ the estimated pose at time t obtained with (21), considering as input $q_{\text{des}}(t)$; ν is a parameter that allows to tune the velocity with which the target pose is reached. It is worth stressing that the objective function minimized by the controller is not linear w.r.t. $\boldsymbol{q}$. As a consequence, the performance of this simple gradient-based controller could be limited by the presence of local minima, and not only by inaccuracies of the GPR model.

VI. Experimental Results

In this section, we describe the experiments that we performed to test the effectiveness of our solutions. We considered tests both in simulated and real environments.

A. Empirical Results for General Trajectories

In this experiment, we investigated the effectiveness of the proposed classification algorithm for general input trajectories, performing a Monte Carlo study composed of 64 experiments in a simulated environment. The simulated environment was implemented in MATLAB. Given the robot DH parameters, and the markers poses expressed w.r.t. the links RFs, the observations were computed evaluating iteratively (2) and (3).

In each experiment, we simulated a different kinematic structure (with $n = 6$), obtaining a dataset of the type defined in (1), with $T = 50$. Among the different simulations we let vary several parameters, like the joint type order $\mathcal{S}_Q$, the DH parameters, the relative pose between the camera and the robot, and the markers attachment, namely, $R_{M_i}^{L_j}$ and $\boldsymbol{l}_{M_i}^{L_j}$. We did not consider the presence of noise. Concerning the input trajectories, each joint trajectory is given by the sum of 10 sinusoids, with amplitudes and angular velocities randomly selected in each simulation. The range of the angular velocities and the amplitudes depends on the joint type. In particular, we considered $[0.1, 1]$[rad$/s$], $[0.1, \pi]$[rad] for revolute joints, and $[0.1, 1][m/s]$, $[0.05, 0.2][m]$ for prismatic joints. For each dataset we considered all the possible triplets, i.e., all the pairs of markers and joint signal, for a total number of 4800 triplets, and we checked the feasibility of the conditions defined by Propositions 1, 2, and 3. We assigned class 1 in case those conditions were satisfied, and class 2 otherwise. The triplets were labeled accordingly: When testing Proposition 1 (resp. Propositions 2 and 3), we assigned the triplet (M_1, M_2, q_k) to class 1 if the elements of the triplet were connected and q_k is a prismatic (resp. revolute) joint. Results are reported in Fig. 4(a) in the form of a confusion matrix: They show that the probability of selecting non*fully informative* input trajectories is close to zero, since in each simulation all the triplets have been labeled correctly. Moreover, results confirm that equations defined by Proposition 2 and 3 identify only a necessary condition when considered alone, since several false positives occurred. However, considering them together, as done in Proposition 5, the number of false positives goes to zero, always leading to a correct classification.

B. UR10: Kinematic Structure Classification

In this experiment, we tested the effectiveness of our kinematic structure classification algorithm with data acquired in a real setup (Fig. 2). Experiments were performed with a UR10

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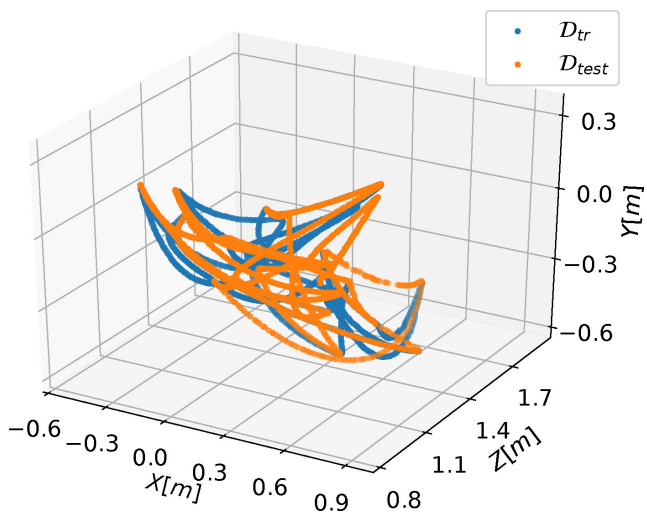

Fig. 5. Plot of the x, y, and z components of $\mathcal{D}_{tr}$ and $\mathcal{D}_{\text{test}}$, expressed in meters. Despite the 25 way-points in common, there are significant differences between the two dataset. The $\mathcal{D}_{tr}$ samples are more widespread toward the bottom-left part of the workspace, while the $\mathcal{D}_{\text{test}}$ samples toward the up-right part.

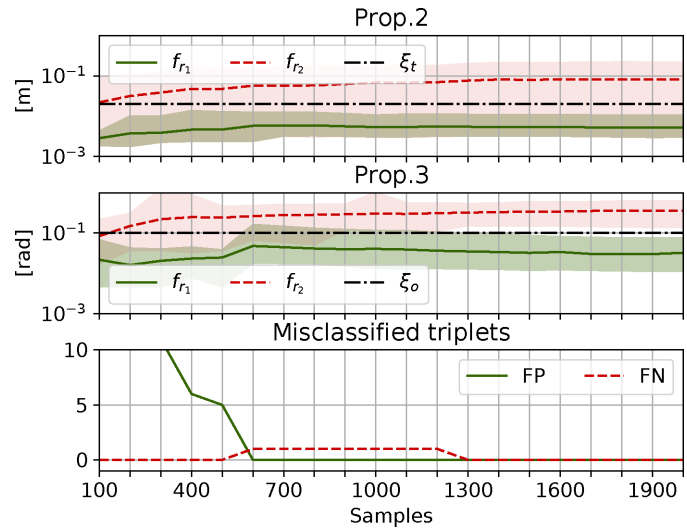

Fig. 6. Evolution of f_{r_1} and f_{r_2}, together with the intervals between the minimum and the maximum values. The bottom plot represents the evolution of the number of *false positives* and *false negatives*, pointed out with FP and FN, respectively.

robot, which is a collaborative manipulator with six revolute joints ($n = 7$), see Fig. 2. As far as the camera is concerned, we used a Microsoft Kinect One (Kinect 2) RGB-D sensor, positioned in front of the robot, at a distance approximately equal to $2[m]$. The AprilTags 36h11 (see [19]) were used as fiducial markers. The interface with the robot and the camera were made with ROS [30], and some ROS packages already available. In particular, we used the URModernDriver [31], which allows collecting the joint positions at $125[Hz]$. While, the frames collected were processed using the IAI Kinect2 ROS package that is a collection of tools and libraries for interfacing the Kinect One with ROS, and the ROS wrapper of the AprilTag 3 visual detector [20], [32], [33]. The real working frequency is limited by the processing time, because the samples acquired when previous data are being processed are neglected. The actual processing time is $1[\text{sec}]$, therefore the whole system works at $1[Hz]$.

We collected two datasets. The first one, hereafter denoted by $\mathcal{D}_{tr}$, consists in the collection of 2103 samples, acquired while the robot follows a trajectory defined by 25 way-points randomly distributed in the available region of the robot workspace. The second dataset, denoted by $\mathcal{D}_{\text{test}}$, is the collection of 2121 samples, collected requiring the robot to reach the same way-points, arranged in a different order; by doing so, the trajectories followed by the robot in the two datasets are different. In Fig. 5 we reported a picture of the X, Y, and Z components of the samples in $\mathcal{D}_{tr}$ and $\mathcal{D}_{\text{test}}$.

As done in the previous experiment, we test all the possible triplets of markers with the observations in $\mathcal{D}_{tr}$. Regarding the values of the thresholds defined in Section IV-C, we set the following values: $\xi_t = 0.02$ and $\xi_o = 0.1$. In Fig. 4(b) we reported the confusion matrix obtained. Results confirm the effectiveness of the strategy proposed to deal with the presence of noise, given that all the triplets are classified correctly.

To check how the number of observations influence the performance of the proposed approach, we analyzed the evolution of the f_r obtained solving the optimization problems in (17) and (18) as function of the training samples. In Fig. 6, we plotted the evolution of the number of misclassified triplets, together with f_{r_1} and f_{r_2}, where f_{r_1} (resp. f_{r_2}) corresponds to the mean of the f_r values associated to triplets that are connected (resp. not connected). In transparency we plotted the intervals between the minimum and the maximum value of f_{r_1} and f_{r_2}. In the initial phases the values of f_{r_1} and f_{r_2} are close, and, consequently, several triplets are not classified correctly. We argue that, due to the small variability of the observations, the residuals due to false assumptions are not significantly different from the residuals due to the noise. The values of f_{r_2} increase significantly with the number of samples, i.e., with the variability of the observations, allowing the correct classification of all the triplets. In particular, in the test performed, the gap between the values of f_{r_1} and f_{r_2} become sufficiently different after 1300 samples. It is worth mentioning that the number of samples required is strictly connected to several aspects, in particular, the variability of the joints trajectories and the measurement noise.

C. UR10: Forward Kinematics Identification

In this experiment, we tested our forward kinematics identification algorithm with data acquired in the real setup. We trained the proposed estimator using the observations in $\mathcal{D}_{tr}$. To test generalization, data efficiency, and robustness w.r.t. the training data, we varied the number of samples, as well as the selection of the samples. Concerning the number of samples selected, we considered $T = 50, 100, 200, 400, 800$. For each value of T, we considered 20 different selections, i.e., 20 different subset of $\mathcal{D}_{tr}$ composed of T samples. The training of each estimator consists in computing (22), after optimizing the hyperparameters by marginal likelihood maximization.

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TABLE I
RMSEs OBTAINED IN $\mathcal{D}_{\text{TEST}}$. RMSEs OF x, y, AND z ARE EXPRESSED IN MILLIMETERS, WHILE γ, β, AND α IN DEGREES

T	$x[mm]$	$y[mm]$	$z[mm]$	$\gamma[deg]$	$\beta[deg]$	$\alpha[deg]$
50	9.53 ±14.84	10.28 ±13.24	14.20 ±15.34	1.32 ±1.06	2.37 ±2.62	2.78 ±3.25
100	4.45 ±3.72	3.18 ±1.82	5.48 ±2.22	0.71 ±0.59	1.21 ±0.40	1.37 ±0.67
200	2.94 ±0.79	1.98 ±0.76	3.65 ±0.93	0.47 ±0.10	0.97 ±0.27	1.01 ±0.17
400	2.55 ±0.12	1.59 ±0.11	2.74 ±0.27	0.38 ±0.03	0.85 ±0.06	0.89 ±0.07
800	2.36 ±0.09	1.46 ±0.07	2.48 ±0.10	0.34 ±0.01	0.80 ±0.03	0.83 ±0.04

TABLE II
RMSEs OF THE POSITIONING ERRORS OBTAINED DERIVING THE CONTROLLER WITH $\mathcal{D}_{tr}$ AND $\bar{\mathcal{D}}_{tr}$. RMSEs OF x, y, AND z ARE EXPRESSED IN MILLIMETERS, WHILE γ, β, AND α IN DEGREES

Dataset	$x[mm]$	$y[mm]$	$z[mm]$	$\gamma[deg]$	$\beta[deg]$	$\alpha[deg]$
$\mathcal{D}_{tr}$	6.51	11.14	22.87	10.61	4.88	6.68
$\mathcal{D}_{tr}$	1.73	1.66	2.18	5.66	1.54	2.40

In Table I, we reported the mean and the three-sigma confidence interval of the root mean squared errors (RMSEs) obtained in $\mathcal{D}_{\text{test}}$, for each value of T. Results show the effectiveness of the proposed solution. The RMSEs of the x, y, and z components are in the order of millimeters, while the RMSEs of the relative orientation are smaller than $1[\text{deg}]$. These values are comparable with the precision of the measurement system. Concerning the dependencies on the number of samples and the data-selection, significant variations can be seen when the number of samples is smaller than 200, which correspond approximately to the 10% of the training data available. This fact denotes good data efficiency.

D. UR10: Control Performance

In this set of experiments, we tested the kinematic controller reported in (25), and based on the model learned with our polynomial-based approach. To derive the estimator, we used all the observations in $\mathcal{D}_{tr}$. Given a fixed initial configuration of the robot, we computed offline the reference trajectories to be provided to the low-level controller to reach five target poses randomly distributed in the robot workspace. After applying the computed control actions, we collected the final poses measured by the camera; the video of the experiment is publicly available.[3] In Table II, we reported the RMSEs of the positioning errors. Results prove that the control actions computed drive the robot close to the target poses. The errors relative to translations are in the order of centimeters, while orientation errors are in the order of a few degrees.

To evaluate the possibility of improving the accuracy of the controller online, we performed a second experiment, in which we derive the forward kinematics estimator using $\bar{\mathcal{D}}_{tr}$. This is the dataset obtained by adding to $\mathcal{D}_{tr}$ the observations of the poses reached with the control actions computed by the previous controller. Results show a consistent improvement, probably due to the higher accuracy of the estimator. This fact highlights the importance of defining strategies to design sufficiently exciting input trajectories.

Finally, we point out that the RMSEs obtained do not depend only on the accuracy of forward kinematic model. Significant contributions to the errors are also due to intrinsic limitations of the control strategy implemented, in particular, the presence of local minima in the error function e_p. These errors might be removed by implementing more complex control strategies, that, however, are out of the scope of this article.

VII. CONCLUSION

In this article, we introduced an autonomous algorithm for kinematic model identification, suitable for robots with serial kinematic composed of revolute and prismatic joints. In the framework presented, we assume, that a distinct marker is attached to each link of the robot, and that joint variables are measurable; no prior knowledge about the robot geometry is assumed. Starting from a time series of such observations, the proposed algorithm first identified the robot kinematic structure, and then, based on the kinematic structure identified, it derived an estimator of the forward kinematics. Such an estimator is based on GPR, in particular on polynomial kernels. We tested the effectiveness of our approach in several simulated environments, as well as in a real setup, with experiments performed with a UR10 robot.

APPENDIX A
PROOF OF PROPOSITION 4

To prove Proposition 4, we introduce the relative transformations between two links L_{j_1} and L_{j_2} which are, in general, not subsequent. The relative orientation between L_{j_1} and L_{j_2}, denoted by $R_{L_{j_2}}^{L_{j_1}}$, is derived iterating (2) along the kinematic chain, namely, considering all transformations due to the joints between L_{j_1} and L_{j_2}, that is

$$R_{L_{j_2}}^{L_{j_1}} = \prod_{m=j_1+1}^{m=j_2} \left(R_z(\theta_m) R_x(\alpha_m)\right) \tag{26}$$

where θ_m is a function of q_{k_m} if the joint is revolute. Instead, as regards $\boldsymbol{l}_{L_{j_2}}^{L_{j_1}}$, i.e., the position of L_{j_2} w.r.t. L_{j_1}, we have

$$\boldsymbol{l}_{L_{j_2}}^{L_{j_1}} = \sum_{m=j_1+1}^{m=j_2} \left(R_{L_{m-1}}^{L_{j_1}} \boldsymbol{l}_{L_m}^{L_{m-1}}\right). \tag{27}$$

Proposition 4 states that, when excitation trajectories are in accordance with Definition 1, if $(M_{i_1} M_{i_2}, q_k)$ verifies equations of Proposition 1 for the minimal set of observations $\bar{\mathcal{D}}$, then L_{j_1} and L_{j_2}, that is, the links to which M_{i_1} and M_{i_2} are attached, are subsequent in the kinematic chain and connected through the prismatic joint q_k.

We prove Proposition 4 by contradiction. First, we consider the case in which there is at least one revolute joint between L_{j_1} and L_{j_2}. Without loss of generality assume that j_3 is the index of the link coming after the revolute joint. The relative orientation between the markers is $R_{M_{i_2}}^{M_{i_1}} = R_{L_{j_1}}^{M_{i_1}} R_{L_{j_3}}^{L_{j_1}} R_{L_{j_2}}^{L_{j_3}} R_{M_{i_2}}^{L_{j_2}}$, which,

[3][Online]. Available: https://youtu.be/PgjvfxMArtQ

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exploiting (26), can be rewritten as

$$R_{M_{i_2}}^{M_{i_1}} = R_{L_{j_1}}^{M_{i_1}} \left(\prod_{m=j_1+1}^{m=j_3-1} R_z(\theta_m) R_x(\alpha_m) \right) \times R_z(\theta_{j_3}) R_x(\alpha_{j_3}) \left(\prod_{\bar{m}=j_3+1}^{\bar{m}=j_2} R_z(\theta_{\bar{m}}) R_x(\alpha_{\bar{m}}) \right) R_{M_{i_2}}^{L_{j_2}}. \tag{28}$$

Now let $\theta_{j_3} = q_{k_{j_3}} + \theta_{j_3}^0$, where $\theta_{j_3}^0$ is the initial value and $q_{k_{j_3}}$ the joint variable. Moreover let us define

$$R_{M_{i_2}}^{\bar{L}_{j_3}} := R_z(\theta_{j_3}^0) R_x(\alpha_{j_3}) \left(\prod_{\bar{m}=j_3+1}^{\bar{m}=j_2} R_z(\theta_{\bar{m}}) R_x(\alpha_{\bar{m}}) \right) R_{M_{i_2}}^{L_{j_2}}.$$

Then it follows:

$$R_{M_{i_2}}^{M_{i_1}} = R_{L_{j_3-1}}^{M_{i_1}} R_z(q_{k_{j_3}}) R_{M_{i_2}}^{\bar{L}_{j_3}}.$$

If conditions of Proposition 1 are verified, then $R_{M_{i_2}}^{M_{i_1}}$ is constant, and, in turn, when considering two different time instants t_1 and t_2, we have $R_{M_{i_2}}^{M_{i_1}}(t_1) = R_{M_{i_2}}^{M_{i_1}}(t_2)$ implying

$$\begin{aligned} &R_{L_{j_3-1}}^{M_{i_1}}(t_1) R_z(q_{k_{j_3}}(t_1)) R_{M_{i_2}}^{\bar{L}_{j_3}}(t_1) \\ &= R_{L_{j_3-1}}^{M_{i_1}}(t_2) R_z(q_{k_{j_3}}(t_2)) R_{M_{i_2}}^{\bar{L}_{j_3}}(t_2). \end{aligned} \tag{29}$$

Now, consider the set of observation $\bar{\mathcal{D}}$ introduced in Definition 1 and, in particular, let us select two input locations belonging to the subset of trajectories with $\text{mod}(q_{k_{j_3}}(t_1)) \neq \text{mod}(q_{k_{j_3}}(t_2))$ and $q_w(t_1) = q_w(t_2)$ if $w \neq k_{j_3}$. Then $R_{L_{j_3-1}}^{M_{i_1}}(t_1) = R_{L_{j_3-1}}^{M_{i_1}}(t_2)$ and $R_{M_{i_2}}^{\bar{L}_{j_3}}(t_1) = R_{M_{i_2}}^{\bar{L}_{j_3}}(t_2)$, but for the uniqueness of the Euler angles $R_z(q_{k_{j_3}}(t_1)) \neq R_z(q_{k_{j_3}}(t_2))$, thus contradicting (29).

To conclude the proof, we consider the case in which there is at least one prismatic joint between L_{j_1} and L_{j_2}. Under this assumption the relative translation between markers is

$$\boldsymbol{l}_{M_{i_2}}^{M_{i_1}} = \boldsymbol{l}_{L_{j_1}}^{M_{i_1}} + R_{L_{j_1}}^{M_{i_1}} \boldsymbol{l}_{L_{j_2}}^{L_{j_1}} + R_{M_{i_2}}^{M_{i_1}} \left(-\boldsymbol{l}_{L_{j_2}}^{M_{i_2}} \right) \tag{30}$$

$$\begin{aligned} &= \sum_{m=j_1+1}^{m=j_2} \left(R_{L_{m-1}}^{M_{i_1}} \left(\begin{bmatrix} 0 \\ 0 \\ d_m \end{bmatrix} + R_z(\theta_m) \begin{bmatrix} a_m \\ 0 \\ 0 \end{bmatrix} \right) \right) \\ &\quad + \boldsymbol{l}_{L_{j_1}}^{M_{i_1}} + R_{M_{i_2}}^{M_{i_1}} \left(-\boldsymbol{l}_{L_{j_2}}^{M_{i_2}} \right) + \sum_{m=j_1+1}^{m=j_2} \left(R_{L_{m-1}}^{M_{i_1}} \begin{bmatrix} 0 \\ 0 \\ q_{k_m} \end{bmatrix} \right) \\ &= \boldsymbol{l}_{i_1*}^{i_2*} + \sum_{m=j_1+1}^{m=j_2} \left(R_{L_{m-1}}^{M_{i_1}} \begin{bmatrix} 0 \\ 0 \\ q_{k_m} \end{bmatrix} \right) \end{aligned} \tag{31}$$

where in $\boldsymbol{l}_{i_1*}^{i_2*}$ we have collected all the terms which are constant, while the terms in the last sum depend on the joint signals. Now, similarly to (7), let us define the parameters vector $\boldsymbol{b}_{i_1*}^{i_2*} = [\boldsymbol{l}_{i_1*}^{i_2*} \ (\boldsymbol{z}_{L_{j_1}}^{M_{i_1}})^T \ \ldots \ (\boldsymbol{z}_{L_{j_2-1}}^{M_{i_1}})^T]$, and the matrix $A_* = [I_3 \ \ q_{k_{j_1}} I_3 \ \ldots \ q_{k_{j_2}} I_3]$, where the $\boldsymbol{z}_{L_{m-1}}^{M_m}$ vectors have unitary norm. Accordingly, let $\mathcal{A}_*(\bar{\mathcal{D}})$ the matrix built stacking the matrices $A_*(t_{j,h})$, $j = 1, 2$, $h = 1, \ldots, n-1$, on top of one another.

If equations of Proposition 1 are verified for the set of observations $\bar{\mathcal{D}}$, then, from (9) it follows that:

$$\boldsymbol{l}\left(P_{i_1}, P_{i_2}\right) = \mathcal{A}(\boldsymbol{q}_k) \boldsymbol{b}_{i_1}^{i_2}$$

but, also

$$\boldsymbol{l}\left(P_{i_1}, P_{i_2}\right) = \mathcal{A}_*(\bar{\mathcal{D}}) \boldsymbol{b}_{i_1*}^{i_2*}.$$

However, when joints trajectories belong to the class defined in Definition 1, $\text{rank}(\mathcal{A}(\boldsymbol{q}_k)) = 6$ and $\text{rank}(\mathcal{A}^*(\bar{\mathcal{D}})) = 3 + 3(j_2 - j_1)$, and, given the constraints on $\boldsymbol{b}_{i_1}^{i_2}$ and $\boldsymbol{b}_{i_1*}^{i_2*}$, $\text{rank}(A(\bar{\mathcal{D}}) \boldsymbol{b}_{i_1}^{i_2}) \leq \text{rank}(A^*(\bar{\mathcal{D}}) \boldsymbol{b}_{i_1*}^{i_2*})$. Then, it must hold $j_2 = j_1 + 1$. Finally, since $\text{span}([q_{k_i}(t_1) \ldots q_{k_i}(t_{n-1})]) = \text{span}([q_{k_j}(t_1) \ldots q_{k_j}(t_{n-1})])$ if and only of $k_i = k_j$, the equivalence in (A) holds only when considering the joint input signal q_k. This concludes the proof of Proposition 4.

Appendix B
Proof of Proposition 5

We prove the proposition by contradiction. Assume that systems defined by Propositions 2 and 3 admit a solution for the set of observations $\bar{\mathcal{D}}$. Consider the case, where L_{j_1} and L_{j_2} are not consecutive. Let N_{rev} be the number of revolute joints present in the chain between the two links, and let $k_{j_3} \ldots k_{j_{3+N_{rev}}}$ be the corresponding indexes. The relative orientation between M_{i_2} and M_{i_1} is $R_{M_{i_2}}^{M_{i_1}} = R_{L_{j_1}}^{M_{i_1}} R_{L_{j_2}}^{L_{j_1}} R_{M_{i_2}}^{L_{j_2}}$, with $R_{L_{j_2}}^{L_{j_1}}$ as in (26). We rewrite $R_{M_{i_2}}^{M_{i_1}}$ highlighting the contributions of a particular revolute joint, for example joint j_3, obtaining $R_{M_{i_2}}^{M_{i_1}} = R_{L_{j_3-1}}^{M_{i_1}} R_z(q_{k_{j_3}}) R_{M_{i_2}}^{\bar{L}_{j_3}}$. Since the system defined by Proposition 3 holds true, we have $R_{M_{i_2}}^{M_{i_1}} = R_{L_{j_1}}^{M_{i_1}} R_z(q_k) R_{M_{i_2}}^{\bar{L}_{j_2}}$, with $R_{L_{j_1}}^{M_{i_1}}$ and $R_{M_{i_2}}^{\bar{L}_{j_2}}$ corresponding to the solution of the nonlinear system. Then it follows:

$$R_{L_{j_1}}^{M_{i_1}} R_z(q_k) R_{M_{i_2}}^{\bar{L}_{j_2}} = R_{L_{j_3-1}}^{M_{i_1}} R_z(q_{k_{j_3}}) R_{M_{i_2}}^{\bar{L}_{j_3}}.$$

Exploiting the standard properties of the rotation matrices and of the Euler angles, it holds $R_z(q_k) = R_{M_{i_1}}^{L_{j_1}} R_{L_{j_3-1}}^{M_{i_1}} R_z(q_{k_{j_3}}) R_{M_{i_2}}^{\bar{L}_{j_3}} R_{\bar{L}_{j_2}}^{M_{i_2}}$. Rewriting $R_{M_{i_1}}^{L_{j_1}} R_{L_{j_3-1}}^{M_{i_1}}$ and $R_{M_{i_2}}^{\bar{L}_{j_3}} R_{\bar{L}_{j_2}}^{M_{i_2}}$ w.r.t. their Euler angles, and adopting the zxz convention, we obtain

$$\begin{aligned} R_z(q_k) = R_z(\alpha_1) R_x(\beta_1) R_z(\gamma_1) R_z(q_{k_{j_3}}) R_z(\alpha_2) \\ R_x(\beta_2) R_z(\gamma_2) \end{aligned}$$

where $(\alpha_1, \beta_1, \gamma_1)$ refers to $R_{M_{i_1}}^{L_{j_1}} R_{L_{j_3-1}}^{M_{i_1}}$ and $(\alpha_2, \beta_2, \gamma_2)$ refers to $R_{M_{i_2}}^{\bar{L}_{j_3}} R_{\bar{L}_{j_2}}^{M_{i_2}}$. Then

$$R_z(q_k - \alpha_1 - \gamma_2) = R_x(\beta_1) R_z(q_{k_{j_3}} + \gamma_1 + \alpha_2) R_x(\beta_2).$$

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Given the uniqueness of the Euler angles, the last equation holds true if and only if β_1 and β_2 are null, and $q_k - \alpha_1 - \gamma_2 = q_{k_{j_3}} + \gamma_1 + \alpha_2$, namely, $q_k - q_{k_{j_3}} = \alpha_1 + \gamma_2 + \gamma_1 + \alpha_2$. Now, consider two observations at time t_1 and t_2, selected from $\bar{\mathcal{D}}$, satisfying $\text{mod}(q_{k_{j_3}}(t_1)) \neq \text{mod}(q_{k_{j_3}}(t_2))$ and $q_w(t_1) = q_w(t_2)$ if $w \neq k_{j_3}$. With this type of trajectory α_1, β_1, γ_1, α_2, β_2, and γ_2 are constant when t is equal to t_1 and t_2. Then, it follows:

$$q_k(t_1) - q_{k_{j_3}}(t_1) = q_k(t_2) - q_{k_{j_3}}(t_2).$$

Under the excitation assumptions in Definition 1, the last equation holds if and only if $k = k_{j_3}$. Similar considerations can be done when considering the other revolute joints, thus implying that only one revolute joint can be present between L_{j_1} and L_{j_2}, and that the corresponding input must be q_k.

To conclude the proof we need to consider the case in which between L_{j_1} and L_{j_2} there is at least one prismatic joint, in addition to the revolute joint q_k. Let j_3 be the index of the link after the revolute joint, that is, the revolute joint connects L_{j_3-1} and L_{j_3}. Consider the expression of $\boldsymbol{l}_{M_{i_2}}^{M_{i_1}}$ in (4), and let us expand $\boldsymbol{l}_{L_{j_2}}^{L_{j_1}}$ highlighting the prismatic transformations that occurs before, or after, the revolute joint. We have

$$\boldsymbol{l}_{M_{i_2}}^{M_{i_1}} = \boldsymbol{l}_{L_{j_1}}^{M_{i_1}} + R_{M_{i_2}}^{M_{i_1}}(-\boldsymbol{l}_{L_{j_2}}^{M_{i_2}}) + \sum_{m=j_1+1}^{m=j_3-1} R_{L_{j_m}}^{M_{i_1}} \boldsymbol{l}_{L_m}^{L_{m-1}} + R_{Lj_3-1}^{M_{i_1}} \boldsymbol{l}_{L_{j_3}}^{L_{j_3-1}} + R_{M_{i_2}}^{M_{i_1}} \sum_{m=j_3+1}^{m=j_2} R_{L_{j_{m-1}}}^{M_{i_2}} \boldsymbol{l}_{L_m}^{L_{m-1}} \quad (32)$$

where the $R_{L_{j_m}}^{M_{i_1}}$, $R_{L_{j_3-1}}^{M_{i_1}}$, and $R_{L_{j_{m-1}}}^{M_{i_2}}$ matrices are constant. The two sums account for the prismatic transformation happening, respectively, before and after the revolute joint; in each sum, the $\boldsymbol{l}_{L_m}^{L_{m-1}}$ expression is defined in (3), where d_j is a function of a suitable joint signal. The translation due to the revolute joint instead, is represented by $R_{Lj_3-1}^{M_{i_1}} \boldsymbol{l}_{L_{j_3}}^{L_{j_3-1}}$, where $\boldsymbol{l}_{L_{j_3}}^{L_{j_3-1}}$ is defined as in (3), with d_{j_3} constant and θ_{j_3} function of q_k. Equation (32) can be rewritten as a linear expression w.r.t. a suitable vector $\boldsymbol{b}_{i_1*}^{i_2*}$ and the following matrix of coefficients

$$A_* = \left[I_3 \; q_{k_{j_1}} I_3 \; \ldots \; q_{k_{j_3}} I_3 \; q_{k_{j_3+1}} R_{M_2}^{M_1} \; \ldots \; q_{k_{j_2}} R_{M_2}^{M_1} \; R_z(q_k) \right].$$

Moreover, recall that if conditions of Proposition 2 are verified $\boldsymbol{l}(P_{i_1}, P_{i_2}) = \bar{\mathcal{A}}(P_{i_1}, P_{i_2}) \bar{\boldsymbol{b}}_{i_1}^{i_2}$, and then

$$\boldsymbol{l}\,(P_{i_1}, P_{i_2}) = \bar{\mathcal{A}}(P_{i_1}, P_{i_2}) \bar{\boldsymbol{b}}_{i_1}^{i_2} = \mathcal{A}_*(\bar{\mathcal{D}}) \boldsymbol{b}_{i_1*}^{i_2*}.$$

If the input trajectories are in accordance with Definition 1 we have that $\text{rank}(\bar{\mathcal{A}}(P_{i_1}, P_{i_2})) = 6$ while $\text{rank}(\mathcal{A}_*(\bar{\mathcal{D}})) = 6 + N_p$, where N_p is the number of prismatic joints. With considerations similar to the ones done in the proof of the previous proposition, we obtain that $N_p = 0$. Then we have proved that conditions of Proposition 5 are verified if and only if the triplet (M_{i_1}, M_{i_2}, q_k) is connected and q_k is revolute.

Appendix C
Proof of Proposition 6

We prove Proposition 6 by inspection. Given that Proposition 6 assumes the knowledge of the kinematic structure, in the following we assume that $\mathcal{Q}_q = \{q_1, \ldots, q_n\}$, that is, joints are provided in accordance with the order with which they occur in the kinematic chain. The relative rotation between the marker attached to the last link and the camera frame is given by

$$R_M^c(\boldsymbol{q}) = R_{L_1}^c R_{L_n}^{L_1}(\boldsymbol{q}) R_M^{L_n}$$

where $R_{L_1}^c$ and $R_M^{L_n}$ are constant matrices, expressing, respectively, the orientation of the base link RF w.r.t. the camera RF and the relative orientation between the last marker and the RF of the last joint. The expression of the matrix $R_{L_n}^{L_1}(\boldsymbol{q})$ instead, is reported in (26), namely, it is factorized by the matrices $R_{L_m}^{L_{m-1}}$, $m = 2, \ldots, n$, defined as in (2). Observe that the elements of $R_{L_m}^{L_{m-1}}$ are constant if the joint between links L_m and L_{m-1} is prismatic, or linear functions of $\cos(q_{m-1})$ and $\sin(q_{m-1})$ if the joint is revolute. Consequently, $R_M^c(\boldsymbol{q})$ is a polynomial function in $\boldsymbol{q}_c$ and $\boldsymbol{q}_s$, with maximum degree N_r, where each monomial is subject to

$$\deg(q_{c_b}) + \deg(q_{s_b}) \leq 1. \quad (33)$$

To prove that the relative translation between the marker attached to the last link and the camera RF is a polynomial function in the elements of $\boldsymbol{q}_c$, $\boldsymbol{q}_s$, and $\boldsymbol{q}_p$ we follow the same reasoning. The expression of $\boldsymbol{l}_M^c$ is

$$\boldsymbol{l}_M^c = \boldsymbol{l}_{L_1}^c + R_{L_1}^c \boldsymbol{l}_{L_1}^{L_n}(\boldsymbol{q}) + R_{L_n}^c(\boldsymbol{q}) \boldsymbol{l}_M^{L_n}$$

where $\boldsymbol{l}_{L_1}^c$ and $\boldsymbol{l}_M^{L_n}$ are the coordinates of the base-link and the marker attached to the last link expressed, respectively, w.r.t. the camera RF and L_n- RF. From (27), $\boldsymbol{l}_{L_1}^{L_n}(\boldsymbol{q})$ can be factorized as $R_{L_{m-1}}^{L_1}(\boldsymbol{q}) \boldsymbol{l}_{L_m}^{L_{m-1}}(\boldsymbol{q})$, where $\boldsymbol{l}_{L_m}^{L_{m-1}}$ is defined as in (3). Its elements are linear functions of $\cos(q_{m-1})$, $\sin(q_{m-1})$ when q_{m-1} is revolute, or of q_{m-1} if q_{m-1} is prismatic. Then, the elements of $R_{L_{m-1}}^{L_1}(\boldsymbol{q})$ are polynomial functions in q_{c_b}, q_{s_b}, with $b = 1, \ldots, m-2$, subject to (33). It turns out that $\boldsymbol{l}_M^c$ is a polynomial function in the elements of $\boldsymbol{q}_c$, $\boldsymbol{q}_s$, and $\boldsymbol{q}_p$, where each element appears with maximum degree 1, and each monomial is subject to the constraint in (33).

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